The Value of a Hospital Obstetric Unit: Access, Costs, and Health*

Thomas Yu[†]

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Abstract

Rates of OB provision, hospitals' decisions to operate obstetric (OB) units, are lower among rural hospitals than urban hospitals and declining overall. I study how OB provision affects access to care, the costs of delivering care, and maternal and infant health. Using data on births and medical claims, I estimate a model of the hospital OB care market in Arkansas. I find that OB provision benefits mothers substantially by giving them access to more options and shortening travel distances, while its effects on health and variable costs are small and mixed in sign. The average OB unit generates a positive externality worth $104 per Arkansas birth, mostly driven by access to care. I evaluate an Arkansas policy that raises Medicaid prices, which keeps OB units open when their costs go up. The policy's benefit is 1.0 to 1.5 times its cost. Alternatively, a flat subsidy to OB hospitals achieves the same goal more cost-effectively, with a benefit 1.2 to 4.2 times its cost.

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[†]Yale University, t.yu@yale.edu

1 Introduction

Obstetric (OB) care, medical care related to pregnancy and childbirth, is primarily delivered in hospital OB units. Mothers in rural parts of the United States often must travel great distances to reach hospitals with OB units. The average distance to the nearest OB hospital is 17.3 miles for birthing mothers living in rural counties, as opposed to 7.3 miles for those living in urban counties (Fontenot et al., 2024).

This difference is partially explained by the fact that even if a rural mother has access to a hospital, it may not have an OB unit. I study OB provision, hospitals' decisions to operate OB units. In 2010, 57% of hospitals in rural counties had OB units, as opposed to 70% of hospitals in urban counties; by 2022, these figures dropped to 48% and 64%, respectively (Kozhimannil et al., 2024). In other words, OB provision rates are lower among rural hospitals than urban hospitals by an increasingly wide margin, and they are declining overall as OB units close.

Policymakers have responded with efforts to keep OB units open. For example, Arkansas's 2025 Healthy Moms, Healthy Babies Act raises the prices paid by Medicaid for OB care by 70% (Arkansas Governor, 2025). The merit of such a policy depends on the social value that OB units provide. In particular, it depends on their net externalities, or how much of that social value is not internalized by OB hospitals themselves.

Previous research on OB unit closures has found mixed results about the effects of OB provision on maternal and infant health (Avdic et al., 2024; Battaglia, 2025; Fischer et al., 2024; Kozhimannil et al., 2018). However, the social value of an OB unit includes not only its effect on health, but also on access and costs. In this paper, I answer the following questions. How does OB provision affect access to care, the costs of delivering care, and maternal and infant health in equilibrium? What are the net externalities of OB provision? What are the consequences of policies aimed at preventing OB unit closures? Focusing on the setting of Arkansas, I quantify the effects of each hospital's OB provision. Then, I evaluate two policies, the Healthy Moms, Healthy Babies Act and an OB unit subsidy.

In a simple example, I clarify four possible welfare effects of OB provision. First, the *access-to-care effect* (giving mothers access to an additional option) increases social welfare. Second, the *fixed cost effect* (fixed cost of operation paid by the OB hospital) decreases social welfare. Third, the *variable cost effect* (variable costs paid by the OB hospital instead of its competitors) has an ambiguous sign depending on the OB hospital's relative marginal cost. Fourth, the *maternal and infant health effect* (the value of better or worse health) has an ambiguous sign depending on the OB hospital's relative health-based quality.

The OB hospital does not fully internalize its effects on mothers, infants, or other hospitals. The net externality of OB provision may be positive or negative, so an unregulated market may

have too little or too much OB provision relative to the welfare-maximizing outcome.

For empirical analysis, I combine two datasets from the Arkansas All-Payer Claims Database. The first contains characteristics of all Arkansas births in 2022, including maternal and infant health. The second contains information from medical claims for commercially-insured and Medicaid-insured Arkansas residents in 2022. I supplement these datasets with American Hospital Association data on hospitals' characteristics, Census Bureau data on demographics, and Centers for Medicare & Medicaid Services data on hospital quality, regional wages, and supplemental payments.

About half of births are commercial, and almost all of the rest are Medicaid. The average distance traveled is 17 miles. The average payment is $8,400 (with $1,200 paid by mothers out-of-pocket) per commercial birth and $3,000 (with zero out-of-pocket spending) per Medicaid birth. Maternal and infant health is defined as an indicator for no adverse outcomes, either maternal (perineal laceration, blood transfusion, unplanned hysterectomy, or uterine rupture) or infant (5-minute Apgar score less than 4, death, injury, or seizure). 3% of births experience any adverse outcome.

I present two sets of descriptive evidence. First, OB hospital choice is sensitive to travel distance: half of mothers choose their nearest option. Second, using variation in mothers' home locations and choice sets, I find no nearest-distance effect, but notable variation in hospital-specific nearest-option effects, on maternal and infant health. Together, this evidence suggests that mothers value shorter travel distances, but whether an OB unit improves or worsens health depends on its quality rather than proximity.

Then, I estimate a model of the hospital OB care market. In the baseline model, commercial prices are set by bargaining between commercial insurers and hospital systems, mothers choose OB hospitals to receive care from, and maternal and infant health is determined. I use a pure characteristics model of OB hospital demand, which has two advantages: it avoids overstating the value of larger choice sets, and it allows me to account for selection by modeling correlations between unobserved preferences and unobserved underlying health. Selection parameters are identified by exogenous choice shifters, most notably home location.

I find that mothers strongly prefer nearby hospitals, with a median value equivalent to $226 more out-of-pocket spending per mile of travel distance. However, the health effect of distance is small: one more mile lowers the probability of no adverse outcomes by 0.02 percentage points. In a cross-hospital comparison, I find that health-based quality explains only 1% of variation in demand-based quality, suggesting that mothers are not fully aware of health effects when choosing OB hospitals. This motivates me to consider maternal and infant health as its own component of social welfare not captured by mothers' demand-based welfare.

Using a Nash-in-Nash bargaining framework, I find that hospitals earn large markups:

commercial births have average marginal costs of $4,800 and markups of $3,600, and Medicaid births have average marginal costs of $1,200 (adjusted downward for supplemental payments) and markups of $1,800.

I apply the estimated model to quantify the effects of OB provision by simulating equilibrium outcomes if each OB unit were to close. I use an estimate of the social cost of adverse outcomes from the literature, and I separate the access-to-care, variable cost, and maternal and infant health effects. The average OB unit's access-to-care effect is worth $162 per Arkansas birth, reflecting the high value that mothers place on access to more options and shorter travel distances. Variable cost effects (on average, $4 per Arkansas birth) and maternal and infant health effects (on average, $8 per Arkansas birth) are much smaller and mixed in sign. Hospitals that are lower-cost (or higher-quality) than their competitors can reduce variable costs (or improve health) by operating OB units; however, higher-cost (or lower-quality) hospitals increase variable costs (or worsen health) by doing so.

I compute net externalities by subtracting private surplus from social surplus. The average OB unit generates a positive externality worth $104 per Arkansas birth, mostly driven by access to care. Mothers are willing to pay a lot for access to OB units, but OB hospitals do not internalize much of this value.

Next, I extend the model to include hospitals' OB provision decisions and estimate the fixed cost of OB provision. I use revealed-preference bounds to set-identify expected fixed cost, which I find to be between $0.35 and 2.21 million.

Finally, I evaluate policy. I begin with the Healthy Moms, Healthy Babies Act's 70% Medicaid price hike. I simulate realistic conditions that may induce OB unit closures: $100,000, $200,000, and $300,000 increases in fixed costs. Across the three scenarios, 2 to 10 of the 37 OB units close without any policy. The Medicaid price hike prevents many closures and improves social welfare by $1 to 16 million, primarily through access to care. The policy costs $29 million in public spending. Its marginal value of public funds ranges from 1.0 to 1.5: each dollar of public spending is a dollar transferred to hospitals plus $0 to $0.50 in additional social welfare.

However, a drawback of the Medicaid price hike is that it pays larger amounts to OB hospitals that attract more Medicaid births, which would have operated OB units anyway. As an alternative policy, I consider a flat subsidy that pays all OB hospitals the same amount. The subsidy also keeps OB units open and improves social welfare (by $1 to 19 million), but with much less public spending ($4 to 11 million). The subsidy's marginal value of public funds ranges from 1.2 to 4.2, achieving the same goal as the Healthy Moms, Healthy Babies Act but doing so more cost-effectively.

Related Literature First, this paper is related to the literature on the effects of hospital and OB unit closures. Some papers use demand and supply models to quantify the welfare effects of closures; for example, Capps et al. (2010) study five hospital closures and find increases in social welfare. Other papers use difference-in-differences strategies to quantify health effects with mixed findings. Joynt et al. (2015) and Petek (2022) find no effects, while Carroll (2026) finds increases in mortality. Focusing on OB units, Kozhimannil et al. (2018) find that OB unit closures lead to more out-of-hospital and preterm births, Avdic et al. (2024) find mixed effects, and Battaglia (2025) and Fischer et al. (2024) find negligible to slightly positive effects on maternal and infant health. These papers are informative about the effects of market participation, but only among providers that actually closed and, for those using difference-in-differences, only on average. I estimate hospital-specific effects of OB provision among all OB hospitals, and I combine health with other components of welfare. I also model OB provision decisions, allowing me to evaluate policies aimed at preventing OB unit closures.

Second, this paper is related to the literature on optimal firm entry. A well-established theoretical literature shows that an unregulated market may have too few or too many participating firms relative to the welfare-maximizing outcome (Mankiw & Whinston, 1986; Spence, 1976). Some empirical papers quantify the welfare effects of market participation; for example, Berry et al. (2016) find too many participants in the local radio market, while Cutler et al. (2010) find that more participants improve welfare when inputs are scarce. I consider a new market (hospital OB care), include health as another component of welfare affected by market participation, and find a positive net externality of market participation.

Third, this paper is related to the literature on health care provider quality estimation. A common issue when estimating provider quality is selection: provider choice may be correlated with unobserved underlying health. A popular strategy is to use patients' home locations as exogenous choice shifters and assume travel distance does not affect health (Einav et al., 2025; Geweke et al., 2003; Olenski & Sacher, 2024). In my model, I allow for distance to affect health and show that home location is still a valid choice shifter because the key variation is in choice sets rather than distances. In addition, out-of-pocket spending and insurance network are two more choice shifters. I consider new applications of provider quality estimates: quantifying the effects of providers' market participation and evaluating policies that affect market participation among providers of varying quality.

Fourth, this paper is related to the literature on models of hospital markets. My model is most similar to that of Gowrisankaran et al. (2015) in which insurers and networks are taken as exogenous and commercial prices are set by bargaining between insurers and hospital systems. I adapt the model to one service line, hospital OB care, and extend it by adding maternal and infant health and OB provision decisions.

Fifth, this paper is related to the literature on pure characteristics demand models. In my model of OB hospital demand, I omit an unobserved consumer-product-level utility shock that often appears in discrete choice models. One reason to do so is that demand models with consumer-product-level utility shocks tend to overstate the value of larger choice sets, which is central to my research question (Ackerberg & Rysman, 2005; Berry & Pakes, 2007). Another reason is that a pure characteristics model makes it feasible to allow all unobserved preferences to correlate with unobserved underlying health, capturing any possible selection. I use methods from Cuesta and Tebaldi (2025), who show that rich individual-level data can help overcome a pure characteristics model's computational challenges.

Sixth, this paper is related to the literature on fixed cost estimation. Because firms that participate in a market may have lower draws of unobserved fixed cost shocks than non-participants, fixed cost estimation cannot rely on expectations conditional on participation. I follow Eizenberg (2014) in assuming worst-case fixed cost bounds to construct expectations unconditional on participation. I apply this strategy to estimate the fixed cost of OB provision, and I use the estimate to evaluate policy.

Outline In Section 2, I describe features of the empirical setting of Arkansas, including the data I use and descriptive evidence. In Section 3, I present a simple example that illustrates the effects of OB provision. In Section 4, I develop an empirical model of demand, health, and prices in the hospital OB care market and discuss identification, estimation, and results. In Section 5, I use the estimated model to quantify the effects of OB provision. In Section 6, I extend the model to include hospitals' OB provision decisions and estimate the fixed cost of OB provision. In Section 7, I evaluate two policies, Arkansas's Healthy Moms, Healthy Babies Act and an OB unit subsidy. In Section 8, I conclude.

2 Arkansas in 2022: Setting, Data, and Descriptives

2.1 Setting

The empirical setting of this paper is Arkansas in 2022. Arkansas is highly rural: among its population of 3 million people, 45% live in rural areas, more than double the national average (Census Bureau, 2024). Arkansas also faces significant maternal and infant health challenges. Its maternal mortality rate (0.35 per thousand births) is 90% higher than the national average, and its infant mortality rate (8.2 per thousand births) is 46% higher than the national average (March of Dimes, 2025). This highlights the importance of health effects.

I exclude three counties because they are parts of metropolitan areas with population centers

in other states: Crittenden County (part of the Memphis metropolitan area) and Miller and Little River Counties (parts of the Texarkana metropolitan area). I exclude specialty hospitals (e.g., psychiatric, rehabilitation, or long-term acute care hospitals) because they do not provide OB care. I exclude hospitals in the Veterans Affairs (VA) system because they serve a particular population that is mostly missing from the claims data I use.

The setting contains 71 hospitals, 37 with OB units and 34 without. Figure 1 shows the locations of these hospitals along with population density. Although Arkansas's overall density is low, it has a few relatively dense urban areas: most notably, the Central Arkansas (Little Rock) and Northwest Arkansas (Fayetteville–Springdale–Rogers) metropolitan areas. OB hospitals are clustered in urban areas but sparse in rural areas.

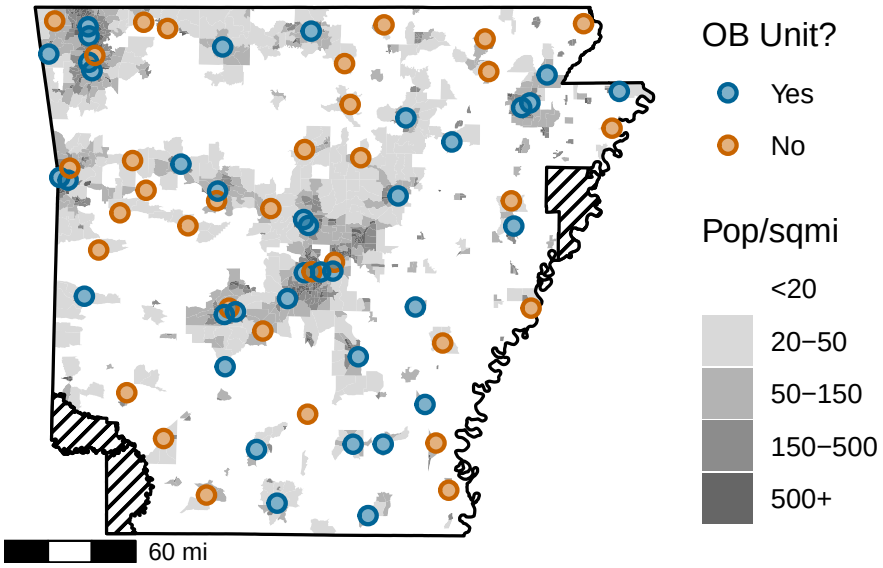


Figure 1: OB Provision in Arkansas in 2022

Data sources: AHA, Census. Sample: hospitals in Arkansas in 2022 (except Crittenden/Miller/Little River Counties and specialty/VA hospitals). Pop densities by census block group. Hatched areas are excluded counties.

Of the 37 OB hospitals, 9 have neonatal intensive care units (NICUs). One OB hospital, the University of Arkansas for Medical Sciences (UAMS), is a teaching hospital with an obstetrics and gynecology residency program, and another, Willow Creek Women's Hospital (WCWH), specializes in medical care for women. These hospitals may have particular demand-based quality, health-based quality, or marginal costs. Furthermore, 37 of the 71 hospitals belong to multi-hospital systems, and together they are owned by 9 multi-hospital systems. OB provision and commercial price-setting decisions are made at the system level.

OB unit closures are especially salient in this setting. In Appendix B.1, I show the history of OB provision. Between 1995 and 2024, there were 24 closures of OB units and zero openings of

OB units in existing hospitals.

In 2025, a policy response to these closures came in the form of the Healthy Moms, Healthy Babies Act. The Arkansas policy invests $45 million per year in measures to improve maternal and infant health; more than half of the investment, $26 million, is dedicated to raising the prices paid by Medicaid for OB care by 70% (Arkansas Governor, 2025). The policy's goal is to prevent OB unit closures, or encourage OB provision, by making it more profitable to deliver OB care. In Section 7, I evaluate this Medicaid price hike along with an alternative policy, a flat subsidy to all OB hospitals.

2.2 Data

I use data from the Arkansas All-Payer Claims Database, as provided by the Arkansas Center for Health Improvement. I combine two datasets. The first contains characteristics of all Arkansas births in 2022, including mothers' characteristics, chosen OB hospitals, and maternal and infant health. The second contains information from medical claims for commercially-insured and Medicaid-insured Arkansas residents in 2022, including payments, out-of-pocket spending, insurers, and networks. In Appendix A.1, I detail how the main data are constructed.

I use six additional sources of data. First, the 2022 Annual Survey from the American Hospital Association (AHA) provides data on hospitals' characteristics, including for-profit status, beds, and NICUs. Second, the 2022 Provider Data Catalog from the Centers for Medicare & Medicaid Services (CMS) provides data on hospital quality, including reviews from Medicare patient surveys, Medicare mortality rates, and Medicare readmission rates. Third, the 2022 CMS Wage Index provides a measure of regional hospital worker wages. Fourth, CMS's Medicaid Budget and Expenditure System provides data on hospitals' supplemental payments for treating Medicaid patients. Fifth, the Census Bureau's 2018–2022 American Community Survey provides data on regional demographics. Sixth, the Census Bureau's 2020 Delineation Files define counties as urban (metropolitan) or rural (not metropolitan).

There are 32,307 births in the sample, and their summary statistics are shown in Table 1. 56% are commercial, 41% are Medicaid, and 3% are neither (e.g., uninsured). Commercial and Medicaid births have similar average distances traveled (17 miles) and adverse outcome rates (3%), where adverse outcomes are either maternal (perineal laceration, blood transfusion, unplanned hysterectomy, or uterine rupture) or infant (5-minute Apgar score less than 4, death, injury, or seizure).

I match 53% of commercial births and 79% of Medicaid births to claims data. There are two reasons why births may be missing claims data. First, some commercial insurers are not required to report to the All-Payer Claims Database: employer-sponsored self-insured plans

Table 1: Birth Summary Statistics

	Commercial	Medicaid	Other
Mean			
Distance (mi)	16.6	16.8	13.8
Any Adverse Outcome	0.03	0.03	0.03
Matched to Claims	0.53	0.79	0
Payment ($)	8399	2962	–
Out-of-pocket ($)	1180	0	–
ARHOME	0.50	0	0
Age (yr)	28.6	26.0	28.7
BMI	28.8	29.0	27.3
Prior Births>0	0.57	0.67	0.69
Prior C-sections>0	0.14	0.16	0.09
Rural	0.29	0.38	0.28
Prenatal Visits	11.0	9.3	9.0
Gestational Age (wk)	38.2	38.1	38.8
Planned C-section	0.22	0.22	0.13
Outside Option	0.009	0.009	0.327
UAMS (Teaching)	0.070	0.132	0.051
WCWH (Women's)	0.051	0.067	0.146
Public	0.08	0.16	0.07
Non-profit	0.74	0.63	0.40
For-profit	0.17	0.20	0.21
NICU	0.57	0.51	0.47
Non-OB Beds	301.8	251.5	155.1
Observations	18003	13360	944

Adverse outcomes: maternal (perineal laceration, blood transfusion, unplanned hysterectomy, uterine rupture) or infant (Apgar<4, death, injury, seizure). Additional summary statistics in Appendix B.2.

and plans with fewer than 2,000 enrollees. Second, there may be ambiguity in mothers' names (which are encrypted) or other data collection errors.

I use births matched to claims data as the main estimation sample. When I compute aggregate measures (e.g., variable costs or profits), I assume commercial births with claims data are similar to those missing claims data, and Medicaid births with claims data are similar to those missing claims data. In other words, I re-weight commercial and Medicaid births to account for their rates of missing claims data. I include the few births covered by neither commercial insurance nor Medicaid in the Medicaid category.[1]

In terms of payments, commercial births have higher payments ($8,400 on average) with a

[1] I assume they are more similar to Medicaid. For many of these births, either the mother is uninsured and would qualify for Medicaid under the generous eligibility rules for pregnant women, or she is insured by another public program (e.g., Medicare or VA).

substantial portion paid by mothers out-of-pocket ($1,200 on average). Medicaid births have lower payments ($3,000 on average) and zero out-of-pocket spending.

In Appendix B.2, I show birth summary statistics split by whether they match to claims data, birth summary statistics by quarter of the year, rates of different adverse outcomes, and hospital-level summary statistics.

I highlight two notes about the data. First, the commercial category includes Medicaid-funded plans. The Arkansas Health and Opportunity for Me (ARHOME) program uses Medicaid funds to buy commercial insurance for residents with incomes between the pre-expansion and expanded Medicaid thresholds. ARHOME plans feature zero out-of-pocket spending like traditional Medicaid, but they feature limited networks and negotiated prices like commercial insurance. In this paper, I classify ARHOME plans as commercial. 50% of the commercial births matched to claims are associated with ARHOME.

Second, distances between mothers and hospitals are imputed expected distances. I observe mothers' home locations at the county level. From the Census Bureau data, I observe populations of women by age bin at the census block group level. I measure the haversine distances between the centroids of census block groups and hospitals. Finally, I use each mother's county and age to construct her probability of living in each census block group and impute her expected distances to hospitals.

2.3 Descriptive Evidence

I present two sets of descriptive evidence that illustrate important features of this setting. First, I show that OB hospital choice is sensitive to travel distance. Figure 2 shows how often mothers choose their nearest, second-nearest, third-nearest, fourth-nearest, and fifth-nearest options in their choice sets, where choice sets are defined as in-network OB hospitals. About half of mothers choose their nearest option, and this pattern holds across different types of mothers. This suggests that travel distance is an important driver of OB hospital choice. It also points to a channel through which OB provision improves mothers' welfare: an additional OB hospital gives some mothers a shorter distance to their nearest options, which, as revealed by their choices, they highly value.

Second, I present evidence about maternal and infant health, suggesting that travel distance is not an important determinant of health while hospital-specific health-based quality is. Let Healthy_i be an indicator for whether mother i experiences no maternal or infant adverse

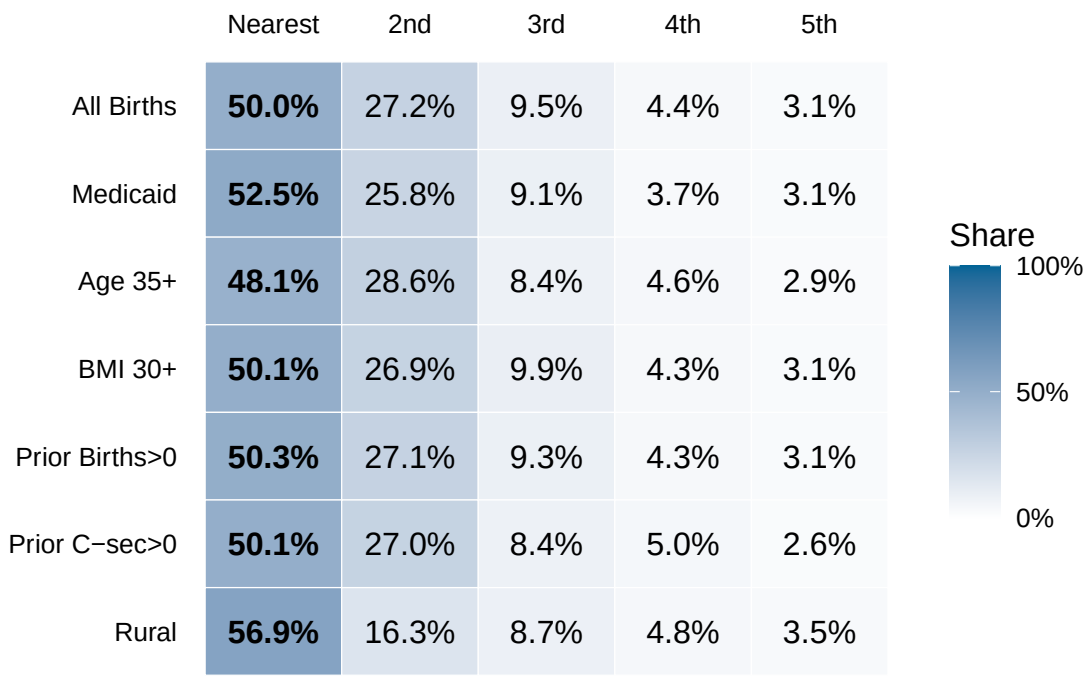


Figure 2: Share of Mothers Who Choose Their Nearest Options

Sample: births with claims data, excluding outside option births. Choice set: in-network OB hospitals.

outcomes. I estimate

$$\text{Healthy}_i = \mathbb{1}\left(\boldsymbol{\tau}'\mathbf{x}_i + \tau^{\text{Dist}}\text{DistNearest}_i + \sum_{j \in J^{\text{OB}}_{m(i)}} \phi_j \mathbb{1}\left(j \text{ nearest to } i\right) + \omega_i > 0\right), \quad \omega_i \sim \text{N}\left(0,1\right) \tag{1}$$

where $\mathbf{x}_i$ consist of a constant, mother's age, mother's race, mother's BMI, a diabetes indicator, a hypertension indicator, a smoker indicator, a prior birth indicator, a prior caesarean section indicator, logged home county population density, and home county population shares by income bin. DistNearest_i is mother i's distance to the nearest OB hospital in her choice set, and $J^{\text{OB}}_{m(i)}$ is her choice set. ω_i is unobserved underlying health. ϕ_j is a hospital-specific nearest-option health effect, and I estimate it as a random effect with $\phi_j \sim \text{N}\left(0, \sigma^2_\phi\right)$.

One parameter of interest is τ^{Dist}, which captures the health effect of distance to the nearest available OB hospital. Another parameter of interest is σ_ϕ, which captures the spread of nearest-option health effects ϕ_j across OB hospitals.

This exercise relies on two key assumptions: mothers tend to choose their nearest option (which is supported by Figure 2), and mothers' home locations are unrelated to their unob-

served underlying health, conditional on observed characteristics.[2] I estimate Equation 1 with maximum likelihood estimation, and Table 2 shows the results. I convert the estimates to average marginal effects, where the nearest-option health effects are relative to a reference hospital, UAMS.

Table 2: Maternal and Infant Health Descriptive Results

	Avg ΔPr(Healthy) (%)
Distance to Nearest Option	
Per Mile	0.013
	(0.019)
Per Standard Deviation (8.6 mi)	0.115
	(0.159)
Nearest-Option Effects	
Standard Deviation Across Hospitals	0.566
	(0.218)
Observations	19843

Results from estimating Equation 1. Sample: births with claims data, excluding outside option births. Drop hospitals that are nobody's nearest option (2 hospitals). Average marginal effects. Reference hospital for nearest-option effects: UAMS. Model-based SEs.

The health effect of distance to the nearest available OB hospital, whether measured per mile or per standard deviation, is near zero and statistically insignificant. This is consistent with the findings of Battaglia (2025) and Fischer et al. (2024), who find no differences in health effects across mothers facing different closure-driven increases in distance to the nearest OB hospital. However, there is notable variation in hospital-specific nearest-option effects. The standard deviation of nearest-option effects across hospitals is 0.57 percentage points with a standard error of 0.22.

On the one hand, despite the sensitivity of mothers' choices to travel distance, its health effect is small. On the other hand, hospital-specific health-based quality may be important. This evidence suggests that whether an OB unit improves or worsens maternal and infant health depends on its health-based quality relative to its competitors rather than how much it reduces travel distances.

3 Simple Example of OB Provision

In this section, I present a simple example that illustrates the effects of OB provision. There is a continuum of mothers of measure 1 distributed uniformly between two hospitals. At one

[2] I support this assumption with descriptive evidence in Appendix B.3.

end, Hospital A operates an OB unit. At the other end, Hospital B may or may not operate an OB unit. An OB unit has fixed cost $FC > 0$. Hospital A's OB unit has constant marginal cost $MC_A > 0$, and Hospital B's OB unit (if it has one) has constant marginal cost $MC_B > 0$. The price of hospital OB care is $p > \max\{MC_A, MC_B\}$.

Each mother chooses to receive care from her preferred OB hospital, and there is no outside option. Mothers incur costs from price and travel distance.[3] Mothers value maternal and infant health, which improves with higher OB hospitals' health-based quality and may also depend on travel distance. I allow for imperfect information by assuming mothers have the following beliefs about health (which may or may not be correct): OB hospitals have the same health-based quality, and travel distance does not affect health. Thus, although mothers are affected by differences in maternal and infant health, they are not aware of these differences when making choices and simply choose their nearest OB hospitals.

At first, suppose only Hospital A operates an OB unit. All mothers choose Hospital A. Because prices are transfers, social welfare consists of aggregate travel costs, aggregate fixed and variable costs, and the value of aggregate maternal and infant health. In other words, social welfare is

$$\text{Welfare}_{\text{A only}} = -\text{TravelCosts}_{\text{A only}} - \text{FC} - \text{MC}_A + \text{Health}_{\text{A only}} \tag{2}$$

Then, suppose Hospital B operates an OB unit as well. The half of mothers located nearer to Hospital B than to Hospital A switch to choosing Hospital B. Social welfare is

$$\text{Welfare}_{\text{A and B}} = -\text{TravelCosts}_{\text{A and B}} - 2\text{FC} - (\text{MC}_A + \text{MC}_B)/2 + \text{Health}_{\text{A and B}} \tag{3}$$

The change in social welfare due to Hospital B's OB provision is

$$\Delta\text{Welfare} = \underbrace{(-\Delta\text{TravelCosts})}_{>0} + \underbrace{(-\text{FC})}_{<0} + \underbrace{(\text{MC}_A - \text{MC}_B)/2}_{>0 \text{ or } <0} + \underbrace{(\Delta\text{Health})}_{>0 \text{ or } <0} \tag{4}$$

where ΔTravelCosts is the change in aggregate travel costs and ΔHealth is the value of the change in aggregate maternal and infant health.

Hospital B's OB provision can affect welfare through four channels. First, the *access-to-care effect* (giving mothers access to an additional option that reduces some of their travel costs) increases social welfare. Second, the *fixed cost effect* (fixed cost paid by Hospital B) decreases social welfare. Third, the *variable cost effect* (variable costs paid by Hospital B instead of Hospital A) has an ambiguous sign depending on Hospital B's marginal cost relative to Hospital A.

Fourth, Hospital B's OB provision can have a *maternal and infant health effect*. Health does

[3] In reality, mothers do not pay the full price because of insurance. In this simple example, mothers' choices can be interpreted as joint mother-insurers' choices, and mothers' welfare as joint mother-insurers' welfare.

not affect mothers' choices due to imperfect information, but it may still affect their welfare. Mothers who switch from Hospital A to Hospital B can experience two health effects: a potential positive effect due to shorter travel distance if care is time-sensitive, and an ambiguously signed effect depending on Hospital B's health-based quality relative to Hospital A. Aggregating over mothers, the health effect may increase or decrease social welfare.

Consider the outcome if a social planner decides whether to operate an OB unit at Hospital B to maximize social welfare. The social planner's OB provision decision is

$$\text{Prov}_{\text{Planner}} = \begin{cases} 1 & \text{if } -\Delta\text{TravelCosts} + (\text{MC}_\text{A} - \text{MC}_\text{B})/2 + \Delta\text{Health} > \text{FC} \\ 0 & \text{if } -\Delta\text{TravelCosts} + (\text{MC}_\text{A} - \text{MC}_\text{B})/2 + \Delta\text{Health} < \text{FC} \end{cases} \tag{5}$$

Alternatively, consider an unregulated market in which Hospital B decides whether to operate an OB unit to maximize its own profit. Let $\pi_\text{B} = (p - \text{MC}_\text{B})/2$ be Hospital B's variable profit upon OB provision. Hospital B's OB provision decision is

$$\text{Prov}_{\text{Private}} = \begin{cases} 1 & \text{if } \pi_\text{B} > \text{FC} \\ 0 & \text{if } \pi_\text{B} < \text{FC} \end{cases} \tag{6}$$

On the one hand, the unregulated market has too little OB provision relative to the welfare-maximizing outcome (Hospital B does not operate an OB unit while the social planner would) if $\pi_\text{B} < \text{FC} < -\Delta\text{TravelCosts} + (\text{MC}_\text{A} - \text{MC}_\text{B})/2 + \Delta\text{Health}$. On the other hand, it has too much OB provision (Hospital B operates an OB unit while the social planner would not) if $-\Delta\text{TravelCosts} + (\text{MC}_\text{A} - \text{MC}_\text{B})/2 + \Delta\text{Health} < \text{FC} < \pi_\text{B}$.

Net Externality Another way to express welfare effects is in terms of an externality. Out of the change in social welfare, Hospital B internalizes its own total profit. Thus, the net externality generated by Hospital B's OB provision decision is

$$\text{Externality} = -\Delta\text{TravelCosts} + (\text{MC}_\text{A} - \text{MC}_\text{B})/2 + \Delta\text{Health} - \pi_\text{B} \tag{7}$$

This suggests an approach to measure the net externality: the OB hospital's total effect on variable social surplus (the sum of the access-to-care, variable cost, and maternal and infant health effects) minus its effect on private surplus (its own variable profit). In Section 5, I use this approach to quantify the net externalities of each hospital's OB provision.

An alternative formulation is

$$\text{Externality} = \underbrace{(-\Delta\text{TravelCosts})}_{>0} + \underbrace{(\text{MC}_\text{A} - p)/2}_{<0} + \underbrace{(\Delta\text{Health})}_{>0 \text{ or } <0} \tag{8}$$

which separates the net externality into three channels: a positive access-to-care externality (giving mothers access to an additional option), a negative business stealing externality (reduction in Hospital A's variable profit), and an ambiguously signed maternal and infant health externality (the value of better or worse health).

If the net externality is positive, it is possible for the unregulated market to have too little OB provision. If the net externality is negative, it is possible for the unregulated market to have too much OB provision.

4 Model of Demand, Health, and Prices

In this section, I develop an empirical model of OB hospital demand, maternal and infant health, and commercial price-setting in the hospital OB care market. I discuss identification, estimation, and results. Hospitals' OB provision decisions are taken as given (in Section 6, I extend the model to include them).

Outcomes are determined in two stages. In Stage 1, demand-based quality, health-based quality, and marginal cost shocks are realized. Then, commercial prices are set by bargaining between commercial insurers and hospital systems. In Stage 2, preference and underlying health shocks are realized. Then, mothers choose OB hospitals to receive care from; simultaneously, maternal and infant health is determined.

I proceed through the stages in reverse order. I begin by discussing Stage 2, OB hospital demand and maternal and infant health. Then, I discuss Stage 1, commercial price-setting. Mothers are indexed by $i \in I$, insurers are indexed by $m \in M$, and hospitals are indexed by $j \in J$. **Bold** characters represent vectors.

4.1 Demand and Health

In Stage 2, preference and underlying health shocks are realized. Then, mothers choose OB hospitals to receive care from; simultaneously, maternal and infant health is determined. Prior to this stage, hospitals' OB provision decisions have been made, demand-based and health-based quality shocks have been realized, and commercial prices have been set.

Each mother is insured by a commercial insurer or Medicaid. Mother i's insurer, denoted

$m(i)$, is taken as exogenous.[4] Let J_m^{OB} denote the set of OB hospitals that are in-network for insurer m. All hospitals are in-network for Medicaid, and networks are taken as exogenous.[5]

Let r_i denote the resource use intensity for mother i's birth, normalized to be 1 on average. Each in-network insurer-OB hospital pair has price p_{mj}. The payment for mother i's birth is $r_i p_{m(i)j}$. In other words, p_{mj} is price per unit of intensity. Given the payment for mother i's birth, she pays $\text{OOP}_i\left(r_i p_{m(i)j}\right)$ out-of-pocket, where the function $\text{OOP}_i(\cdot)$ is determined by her deductible, copay, and coinsurance rate. r_i and $\text{OOP}_i(\cdot)$ are taken as exogenous. In Appendix A.2, I detail how r_i, p_{mj}, and $\text{OOP}_i(\cdot)$ are recovered from the data.

Mothers' home locations are taken as exogenous.[6] Let Dist_{ij} denote the distance between mother i's home and hospital j.

Each mother maximizes her utility by choosing among her choice set: in-network OB hospitals plus an outside option (which includes possibilities such as out-of-network OB hospitals, non-OB hospital emergency rooms, birth centers, or home births).[7] Mother i's utility from choosing OB hospital j is

$$u_{ij} = -\alpha \text{OOP}_i\left(r_i p_{m(i)j}\right) - \beta_i\left(\text{Dist}_{ij} + \eta \text{Dist}_{ij}^2\right) + \boldsymbol{\gamma}_i' \mathbf{y}_j + \xi_j \tag{9}$$

where $\mathbf{y}_j$ are hospitals' characteristics and ξ_j is an unobserved demand-based quality shock. The random coefficients β_i and $\boldsymbol{\gamma}_i$ are parameterized as

$$\beta_i = \exp\left(\delta_0^{\text{Dist}} + \boldsymbol{\delta}^{\text{Dist}\prime} \mathbf{x}_i + \nu_i^{\text{Dist}}\right) \tag{10}$$

$$\gamma_i^k = \delta_0^k + \boldsymbol{\delta}^{k\prime} \mathbf{x}_i + \nu_i^k \quad \text{for all hospital characteristics } k \tag{11}$$

where $\mathbf{x}_i$ are mothers' characteristics and $\boldsymbol{\nu}_i$ are unobserved preference shocks.

Simultaneously, maternal and infant health is determined. Let Healthy_{ij} be an indicator for whether mother i choosing OB hospital j experiences no maternal or infant adverse outcomes. Healthy_{ij} is determined by

$$\text{Healthy}_{ij} = \mathbb{1}\left(\boldsymbol{\zeta}' \mathbf{x}_i + \boldsymbol{\kappa}' \mathbf{y}_j + \boldsymbol{\mu}' \mathbf{z}_{ij} + \lambda_j + \epsilon_i > 0\right) \tag{12}$$

[4] In Appendix B.4, I present descriptive evidence suggesting that mothers do not choose insurers for reasons related to OB care.

[5] Network formation happens at the hospital level and is unlikely to be driven by OB care. The share of revenue from OB care is 3.2% among OB hospitals.

[6] In Appendix B.3, I present descriptive evidence suggesting that conditional on observed characteristics, mothers' home locations are unrelated to OB care.

[7] Almost all mothers deliver at the hospital where their prenatal care provider practices, so the choice of OB hospital is effectively the joint choice of a prenatal care provider and hospital. In Appendix B.5, I argue that this interpretation is consistent with the model.

where $\mathbf{x}_i$ are mothers' characteristics, $\mathbf{y}_j$ are hospitals' characteristics, $\mathbf{z}_{ij}$ are interacted mother-hospital characteristics, λ_j is an unobserved health-based quality shock, and ϵ_i is an unobserved underlying health shock.

I parameterize the joint distributions of unobserved shocks:

$$\begin{pmatrix} \boldsymbol{\nu}_i \\ \epsilon_i \end{pmatrix} \sim \mathrm{N}\left(0, \Sigma_{\nu\epsilon}\right) \tag{13}$$

$$\begin{pmatrix} \xi_j \\ \lambda_j \end{pmatrix} \sim \mathrm{N}\left(0, \Sigma_{\xi\lambda}\right) \tag{14}$$

Choice Probabilities, Expected Utility, and Willingness to Pay Given this structure, the probability that mother i chooses OB hospital j is

$$s_{ij}\left(J^{\mathrm{OB}}_{m(i)}, \mathbf{p}_{m(i)}\right) = \Pr\left(j = \underset{k \in J^{\mathrm{OB}}_{m(i)} \cup \{\text{outside}\}}{\operatorname{argmax}} u_{ik} \;\middle|\; \boldsymbol{\xi}\right) \tag{15}$$

and mother i's expected utility is

$$\mathrm{EU}_i\left(J^{\mathrm{OB}}_{m(i)}, \mathbf{p}_{m(i)}\right) = \mathrm{E}\left[\max_{j \in J^{\mathrm{OB}}_{m(i)} \cup \{\text{outside}\}} u_{ij} \;\middle|\; \boldsymbol{\xi}\right] \tag{16}$$

where uncertainty is over preference shocks $\boldsymbol{\nu}_i$.

Mother i's willingness to pay for a set of in-network OB hospitals J' is

$$\mathrm{WTP}_{iJ'} = \frac{1}{\alpha}\left[\mathrm{EU}_i\left(J^{\mathrm{OB}}_{m(i)}, \mathbf{p}_{m(i)}\right) - \mathrm{EU}_i\left(J^{\mathrm{OB}}_{m(i)} \setminus J', \mathbf{p}_{m(i)}\right)\right] \tag{17}$$

where $\frac{1}{\alpha}$ converts expected utility to monetary units.

Discussion of Demand and Health Model I discuss two notable features of this model. First, I address selection, a common issue in health outcome models: $\mathrm{Healthy}_{ij}$ is only observed for ij pairs where mother i chooses OB hospital j. This can introduce bias if a mother's unobserved preferences are correlated with her unobserved underlying health. For example, if mothers who are unobservably sicker (lower ϵ_i) are unobservably more likely to choose hospitals with characteristic k (higher ν_i^k), then naive estimation of Equation 12 (e.g., simple probit regression) would underestimate κ^k: hospitals with characteristic k would appear lower-quality because they attract sicker mothers.

To account for selection, I leverage the demand model and model correlations between

unobserved preferences and unobserved underlying health in $\Sigma_{\nu\epsilon}$. Then, if sicker mothers are more likely to choose hospitals with characteristic k, the correlation would be captured by $\text{Cov}\left(\nu_i^k, \epsilon_i\right)$ instead of biasing the estimate of κ^k.

Second, I model demand as pure characteristics, omitting from Equation 9 an unobserved ij-level (usually Gumbel-distributed) utility shock that often appears in discrete choice models. Instead, all mother-level unobserved preferences are contained in the preference shocks $\boldsymbol{\nu}_i$ and are thus disciplined by hospitals' characteristics. I do so for two reasons.

One reason is that demand models with ij-level shocks, which imply the characteristic space expands without bound as the number of choices increases, tend to overstate the value of larger choice sets (Ackerberg & Rysman, 2005; Berry & Pakes, 2007). Because the value of larger choice sets is central to my research question, omitting an ij-level shock is a conservative measure. Another reason is that an ij-level shock opens up an additional channel of unobserved preferences, and allowing this channel to correlate with underlying health shocks ϵ_i is infeasible: correlations between ij-level shocks and ϵ_i introduce too many parameters (as many as the number of OB hospitals) to reliably identify given the rarity of adverse outcomes.[8] A pure characteristics model makes it feasible to force all unobserved preferences into a channel that correlates with ϵ_i, capturing any possible selection.

The downside of a pure characteristics model is computational complexity, as an ij-level shock provides helpful smoothing properties. I use methods from Cuesta and Tebaldi (2025), who show that rich individual-level data provide similar smoothing properties that make computation tractable.

Identification of Demand and Health Parameters I normalize $\alpha = 1$ and $\text{Var}\left(\epsilon_i\right) = 1$. The parameters to identify are $\left\{\delta_0^{\text{Dist}}, \boldsymbol{\delta}^{\text{Dist}}\right\}$, $\left\{\delta_0^k, \boldsymbol{\delta}^k\right\}$ for all hospital characteristics k, η, $\boldsymbol{\zeta}$, $\boldsymbol{\kappa}$, $\boldsymbol{\mu}$, and $\Sigma_{\nu\epsilon}$. I also identify $\Sigma_{\xi\lambda}$ and the realized values of $\left\{\xi_j, \lambda_j\right\}$ for all OB hospitals j.

For all mothers i and OB hospitals j, I observe $\text{OOP}_i\left(r_i p_{m(i)j}\right)$, Dist_{ij}, $\mathbf{x}_i$, $\mathbf{y}_j$, and $\mathbf{z}_{ij}$. I also observe networks J_m^{OB} and choices $\mathbb{1}\left(i \text{ chooses } j\right)$. Finally, I observe maternal and infant health at chosen OB hospitals, or $\text{Healthy}_{ij(i)}$.

δ_0^{Dist} is identified by how choice probabilities vary with distance, and $\boldsymbol{\delta}^{\text{Dist}}$ are identified by how that distance preference varies with mothers' characteristics. Similarly, for each hospital characteristic k, δ_0^k is identified by how choice probabilities vary with k, and $\boldsymbol{\delta}^k$ are identified by how that preference for k varies with mothers' characteristics. η is identified by the distance preference's curvature. $\text{Var}\left(\boldsymbol{\nu}_i\right)$ are identified by choices not explained by common preferences. For example, suppose some mothers choose their nearest option even if better options exist

[8] Geweke et al. (2003) estimate such a model using Bayesian inference methods in a setting with a higher adverse outcome rate than the setting of this paper.

slightly farther away, while other mothers travel far to choose their best options; this implies $\text{Var}\left(\nu_i^{\text{Dist}}\right)$ is large.

$\boldsymbol{\zeta}$ are identified by how health varies with mothers' characteristics. Separately identifying the health effects in $\boldsymbol{\kappa}$ and $\boldsymbol{\mu}$ from the selection in $\Sigma_{\nu\epsilon}$ is more difficult because both are related to how health varies with hospitals' characteristics and travel distance. I separate them with exogenous choice shifters, variables that do not directly affect health. The primary choice shifter is home location, and two more choice shifters exist for commercial mothers: out-of-pocket spending and network. Using variation in choice shifters, health effects are identified by how health varies with choice probabilities and expected travel distances. Selection is identified by how health varies with unlikely choices and travel distances.

The key variation is in choice sets.[9] For example, mothers with the same distance to their nearest option but different sets of options have different choice probabilities, so their health differences identify the health effects of hospitals' characteristics. Mothers with the same options, but different distances away, have different expected travel distances, so their health differences identify the health effect of distance. Mothers who choose a less likely hospital, or travel farther than expected, reveal their unobserved preferences, so whether their health is better or worse than expected identifies the selection parameters.

ξ_j is identified by the common preference for hospital j not explained by observed characteristics, and $\text{Var}\left(\xi_j\right)$ is identified by the spread of those preferences. Similarly, λ_j is identified by the common health effect of hospital j not explained by observed characteristics, using the same choice-set variation as $\boldsymbol{\kappa}$, and $\text{Var}\left(\lambda_j\right)$ is identified by the spread of those health effects. $\text{Cov}\left(\xi_j, \lambda_j\right)$ is identified by the correlation between those preferences and health effects.

Estimation of Demand and Health Parameters For mothers' characteristics in Equations 10 and 11, I use a Medicaid indicator, an ARHOME indicator, mother's age, mother's BMI, a prior birth indicator, a prior caesarean section indicator, and logged home county population density. For hospitals' characteristics in Equation 9, I use an outside option indicator, a UAMS indicator, a WCWH indicator, a NICU indicator, logged non-OB beds, and an index constructed using reviews from Medicare patient surveys.

For mothers' characteristics in Equation 12, I use a constant, a Medicaid indicator, an ARHOME indicator, mother's age, mother's race, mother's BMI, a diabetes indicator, a hypertension indicator, a smoker indicator, a prior birth indicator, a prior caesarean section indicator, logged home county population density, and home county population shares by income bin. For hospitals' characteristics, I use an outside option indicator, a UAMS indicator, a NICU

[9] Previous work using home location as a choice shifter (Einav et al., 2025; Geweke et al., 2003; Olenski & Sacher, 2024) assumes travel distance does not affect health. In this paper, the key variation is in choice sets rather than distances, which frees distance to have a health effect of its own.

indicator, logged non-OB beds, and an index constructed using outcomes for Medicare patients (specifically, mortality and readmission rates for pneumonia, COPD, and heart failure). For interacted mother-hospital characteristics, I use a selected set of hospitals' characteristics (UAMS, NICU, logged non-OB beds, and the patient outcomes index) interacted with a selected set of mothers' characteristics (age, BMI, prior birth, and prior caesarean section). I also use distance and distance interacted with those same mothers' characteristics.

In both the demand and health models, all mothers' characteristics are plausibly invariant to the chosen OB hospital. I omit characteristics that may vary for reasons related to hospital choice, such as prenatal care provider, prenatal care visits, gestational age, or a caesarean section indicator.

I assume the outside option has the insurer-level average price. Other than out-of-pocket spending and the outside option indicator, the outside option's characteristics are set to 0. I assume $\delta^{\text{outside}} = \delta^{\text{WCWH}} = 0$ and $\nu_i^{\text{outside}} = \nu_i^{\text{WCWH}} = 0$ for all mothers i; in other words, there is no observed or unobserved heterogeneity in preferences for either the outside option or WCWH. Furthermore, I assume $\text{Cov}(\boldsymbol{\nu}_i)$ is diagonal; in other words, unobserved preferences for different hospital characteristics are independent of each other.

Because common preferences for the outside option, UAMS, and WCWH are entirely captured by the coefficients on their indicators, leaving no room for unobserved components, I normalize ξ_{outside}, ξ_{UAMS}, and ξ_{WCWH} to 0. For the same reason in health effects, I normalize λ_{outside} and λ_{UAMS} to 0.

I apply two steps of maximum likelihood estimation. In the first step, using observed choices, I estimate the demand parameters. I estimate hospital-specific fixed effects, which are each separated by generalized least squares into the component explained by observed characteristics (to obtain δ_0^k for all hospital characteristics k) and the unobserved component (to obtain σ_ξ and ξ_j for all OB hospitals j). In the second step, I take the demand estimates as given, including estimates of ξ_j adjusted with empirical Bayes shrinkage. Using observed choices and health outcomes, I estimate the health parameters, including λ_j as a random effect. Second-step standard errors are corrected for first-step imprecision by drawing demand parameters from their covariance. Table 3 shows the main results, which are related to distance and hospital size. In Appendix C, I present the complete set of results.

Mothers strongly prefer nearby hospitals: the median value that a mother places on an additional mile of travel distance is equivalent to \$226 more out-of-pocket spending.[10] However, the effect of distance on maternal and infant health is small. An additional mile lowers the probability of experiencing no adverse outcomes by 0.02 percentage points. This suggests

[10] I highlight medians instead of averages because distance effects are skewed given the lognormal distribution of β_i. The average value of distance is equivalent to \$940 more out-of-pocket spending per mile.

Table 3: Main Demand and Health Results

	Median ΔUtility ($)	Avg ΔPr(Healthy) (%)
Distance (mi)	−225.86	−0.02138
	(69.91)	(0.00922)
ΔEffect of Distance (mi) per Δx_i:		
Age (yr)	−0.01	−0.00056
	(1.75)	(0.00066)
BMI	1.73	0.00099
	(1.34)	(0.00054)
Prior Births>0	−41.16	0.01006
	(24.77)	(0.00978)
Prior C-sections>0	39.20	0.00004
	(25.33)	(0.01379)
	Avg ΔUtility ($)	Avg ΔPr(Healthy) (%)
2×Non-OB Beds	3638.0	−0.187
	(2198.3)	(0.383)
ΔEffect of 2×Non-OB Beds per Δx_i:		
Age (yr)	119.0	−0.001
	(33.1)	(0.014)
BMI	59.0	0.024
	(15.7)	(0.013)
Prior Births>0	−521.1	0.168
	(156.0)	(0.179)
Prior C-sections>0	633.3	0.123
	(376.9)	(0.288)
Observations	20087	

Sample: births with claims data. Utility effects of distance are median marginal effects evaluated at chosen distances. All other effects are average marginal effects. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation. Additional results in Appendix C.

mothers value proximity for reasons unrelated to health.

Hospital size, measured by the number of non-OB beds, plays a role in demand as well. Mothers prefer larger hospitals: on average, mothers value a hospital with twice as many non-OB beds equivalently to $3,600 less out-of-pocket spending. This preference is stronger for older mothers and mothers with higher BMIs (who may expect to have more complicated deliveries), and it is weaker for mothers with prior births without caesarean sections (who may expect to have less complicated deliveries). In contrast, I find no evidence of a systematic effect of hospital size on maternal and infant health. As with distance, mothers value hospital size for reasons unrelated to health.

Finally, I quantify the extent to which mothers' choices are driven by health in a cross-

hospital comparison. I define each hospital j's demand-based quality as the average mother's increase in utility from choosing hospital j relative to the outside option, setting distance and out-of-pocket spending to zero. In other words, I measure how much mothers would value hospital j if it were in-network, free, and required no travel. Then, I define each hospital j's health-based quality as the average mother's change in no-adverse-outcome rate relative to the outside option, setting distance to zero. In other words, I measure how much hospital j would improve maternal and infant health if it required no travel.

Figure 3 shows the relationship between demand-based and health-based quality. The slope of their relationship is not statistically significant, and health-based quality explains only 1% of variation in demand-based quality. This result is consistent with mothers not being fully aware of differences in maternal and infant health when choosing OB hospitals. In social welfare calculations, I assume maternal and infant health is not captured by mothers' demand-based welfare and include it as its own component of social welfare.

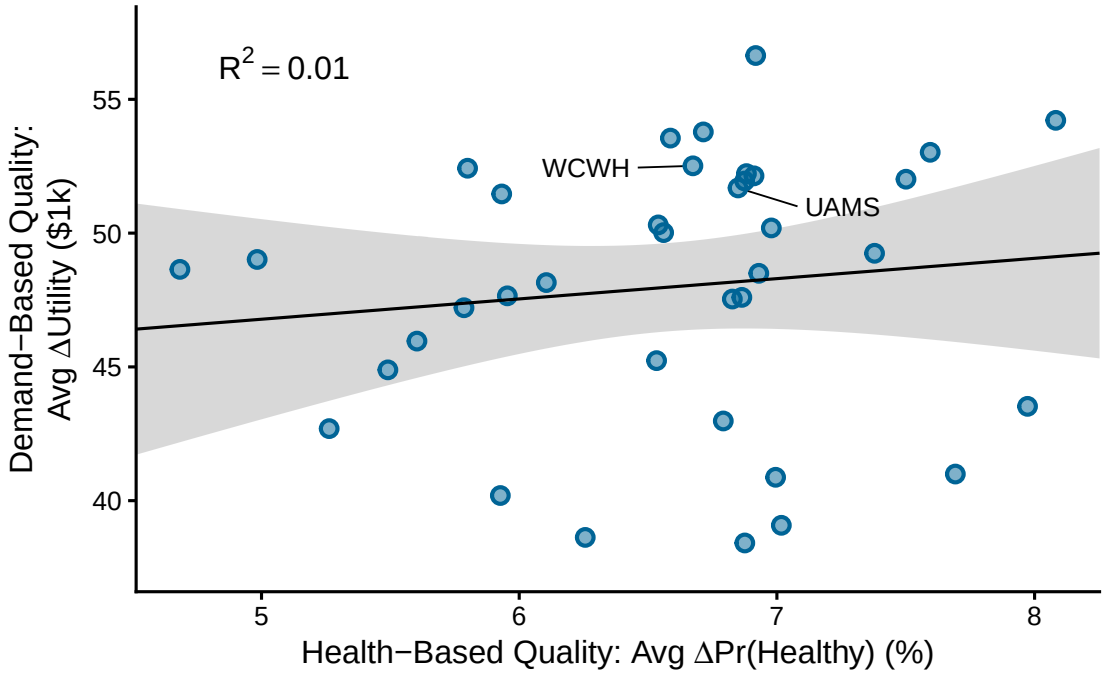


Figure 3: Relationship Between Demand-Based and Health-Based Quality

Sample: OB hospitals. Health-based quality is average change in no-adverse-outcome rate relative to outside option, ignoring distance. Demand-based quality is average change in utility relative to outside option, ignoring distance and OOP. Labeled hospitals are UAMS (teaching) and WCWH (women's). Sloped line from OLS regression. Shaded area is 95% CI, taking quality estimates as given.

4.2 Commercial Price-Setting

In Stage 1, health-based quality, demand-based quality, and commercial marginal cost shocks are realized. Then, commercial prices are set by bargaining between commercial insurers and hospital systems. Prior to this stage, hospitals' OB provision decisions have been made. Insurers and systems have expectations about OB hospital demand, where uncertainty is over unrealized preference shocks $\boldsymbol{\nu}_i$ for all mothers i. Let I_m denote the set of mothers insured by insurer m and J_s^{OB} denote the set of OB hospitals that belong to system s.

Each insurer is a perfect agent for its enrollees: it values its enrollees' expected utility but incurs costs from expected payments net of out-of-pocket spending. Specifically, insurer m's value (as a function of its in-network OB hospitals J_m^{OB} and prices $\mathbf{p}_m$) is

$$V_m\left(J_m^{\text{OB}}, \mathbf{p}_m\right) = \frac{1}{\alpha} \sum_{i \in I_m} \text{EU}_i\left(J_m^{\text{OB}}, \mathbf{p}_m\right) - \sum_{i \in I_m} \sum_{j \in J_m^{\text{OB}}} \left[s_{ij}\left(J_m^{\text{OB}}, \mathbf{p}_m\right)\left(r_i p_{mj} - \text{OOP}_i\left(r_i p_{mj}\right)\right)\right] \quad (18)$$

where $\frac{1}{\alpha}$ converts expected utility to monetary units.

I assume markups (prices net of marginal costs) are constant within an insurer-system pair. If system s is in-network for insurer m, they engage in Nash-in-Nash bargaining over $\text{Markup}_{ms} = p_{mj} - \text{MC}_{mj}$ for all $j \in J_s^{\text{OB}}$. Given all other prices, their markup maximizes the exponentiated product of insurer m's gain in value and system s's gain in expected variable profit from network inclusion. Specifically, Markup_{ms} maximizes

$$\begin{aligned} \text{NB}_{ms}\left(\text{Markup}_{ms} \mid \mathbf{p}_{m,-s}\right) = &\left[V_m\left(J_m^{\text{OB}}, \mathbf{p}_m\right) - V_m\left(J_m^{\text{OB}} \setminus J_s^{\text{OB}}, \mathbf{p}_{m,-s}\right)\right]^b \\ &\left[\sum_{i \in I_m} \sum_{j \in J_s^{\text{OB}}} \left[s_{ij}\left(J_m^{\text{OB}}, \mathbf{p}_m\right) r_i\right] \text{Markup}_{ms}\right]^{1-b} \end{aligned} \quad (19)$$

where $b \in [0, 1]$ is the bargaining weight of insurers relative to hospital systems and MC_{mj} is marginal cost per unit of intensity.

Given this structure, each commercial price p_{mj} satisfies

$$p_{mj} = \text{MC}_{mj} - \left[\frac{\partial Q_{ms}}{\partial \text{Markup}_{ms}} - \left(\frac{b}{1-b}\right)\left(\frac{\partial V_m\left(J_m^{\text{OB}}, \mathbf{p}_m\right) \,/\, \partial \text{Markup}_{ms}}{V_m\left(J_m^{\text{OB}}, \mathbf{p}_m\right) - V_m\left(J_m^{\text{OB}} \setminus J_s^{\text{OB}}, \mathbf{p}_{m,-s}\right)}\right)\right]^{-1} Q_{ms} \quad (20)$$

where $Q_{ms} = \sum_{i \in I_m} \sum_{j \in J_s^{\text{OB}}} \left[s_{ij}\left(J_m^{\text{OB}}, \mathbf{p}_m\right) r_i\right]$.

I further parameterize marginal cost MC_{mj} as

$$\text{MC}_{mj} = \boldsymbol{\psi}' \mathbf{w}_{mj} + e_{mj}, \quad e_{mj} \sim \text{N}\left(0, \sigma_e^2\right) \quad (21)$$

where $\mathbf{w}_{mj}$ are insurer-hospital characteristics and e_{mj} is an unobserved marginal cost shock. I assume e_{mj} is independent of hospital j's demand-based and health-based quality shocks.

Discussion of Commercial Price-Setting Model I discuss two topics related to this model. First, marginal cost includes not only direct accounting costs (e.g., nurses, medications, or supplies), but also indirect costs and benefits internalized by hospital systems. For example, marginal costs may be lowered by public and non-profit hospitals' mission-guided incentives to provide care, or by OB patients consuming other services at the same hospital.

Second, I make simplifying assumptions about the model's bargaining framework. I assume insurers are perfect agents that value enrollees' utility one-for-one with payments, and I assume markups are constant within an insurer-system pair. In the insurer-hospital bargaining literature, insurers are often modeled as imperfect agents with a relative utility weight, and multi-hospital systems often flexibly divide markups among their hospitals (Gowrisankaran et al., 2015). However, these two features, an insurer's utility weight and a system's division of markups, would be identified solely by variation in demand elasticity with respect to price, which is driven by coinsurance. In this setting, there is not enough variation in coinsurance rates (which are zero for most mothers) for reliable identification.

Furthermore, I assume a common relative bargaining weight b for all insurer-system pairs. I do so because cross-insurer and cross-hospital variation in bargaining weights would be difficult to identify alongside variation in marginal costs. I choose to focus on marginal costs, which are more closely related to my research question than bargaining weights are.

Identification of Commercial Price-Setting Parameters The parameters to identify are b, $\boldsymbol{\psi}$, σ_e, and the realized values of e_{mj} for all in-network commercial insurer-OB hospital pairs mj. For all mothers i and OB hospitals j, I observe r_i, $p_{m(i)j}$, $\text{OOP}_i(\cdot)$, and $\mathbf{w}_{m(i)j}$. In addition, $s_{ij}(\cdot)$ and $\text{EU}_i(\cdot)$ are identified from the OB hospital demand stage.

For candidate values of b and $\boldsymbol{\psi}$, Equation 20 gives implied values of marginal costs MC_{mj}, and then Equation 21 gives implied values of marginal cost shocks e_{mj}. The true parameter values uniquely satisfy

$$\text{E}\left[e_{mj}(b, \boldsymbol{\psi})\, \widetilde{\mathbf{w}}_{mj}\right] = 0 \tag{22}$$

where $\widetilde{\mathbf{w}}_{mj}$ are instruments plausibly assumed to be related to price but independent of e_{mj}. The moments in Equation 22 for all in-network commercial insurer-OB hospital pairs identify b, $\boldsymbol{\psi}$, and e_{mj} for all pairs. The spread of e_{mj} identifies σ_e.

$\widetilde{\mathbf{w}}_{mj}$ consist of $\mathbf{w}_{mj}$ along with an additional instrument: for insurer m and hospital j, the average willingness to pay among insurer m's enrollees for hospital j's system if all prices were equal to the average observed price.

The following explanations provide intuition for how the parameters are identified. First, $\boldsymbol{\psi}$ are identified by how price varies with $\mathbf{w}_{mj}$. Second, b is identified by how price varies with the willingness to pay instrument. If b is low, then hospital systems have more bargaining power; more desirable systems can command higher prices, so willingness to pay is positively related to price. If b is high, then insurers have more bargaining power; prices are driven down to marginal costs, so willingness to pay is less closely related to price. Third, marginal cost shocks e_{mj} are identified by price variation not explained by observed characteristics, and σ_e is identified by their spread.

Estimation of Commercial Price-Setting Parameters For insurer-hospital characteristics in Equation 21, I use a constant, a UAMS indicator, a NICU indicator, logged non-OB beds, a for-profit indicator, the CMS Wage Index (which measures regional hospital worker wages), and an indicator for ARHOME. When computing gains from network inclusion, I assume the price of the outside option is the insurer-level average price.

Using empirical analogs of the moments in Equation 22, I apply generalized method of moments. Standard errors are clustered by insurer and system, and they take demand estimates as given. Table 4 shows the results.

Births at UAMS, the only teaching hospital with an obstetrics and gynecology residency program, cost \$900 more per unit of intensity than births at otherwise identical hospitals. Medicaid-funded ARHOME births cost \$1,300 less per unit of intensity than other commercial births. Insurers' relative bargaining weight is 0.6, suggesting that insurers capture slightly more than half of the surplus from network inclusion, although I cannot reject an even split. The estimates imply hospitals earn large markups. Among non-outside-option commercial births, the average marginal cost is \$4,790, and the average markup is \$3,614.

Medicaid Prices and Marginal Costs Counterfactual simulations require estimates of Medicaid prices and marginal costs. For OB hospitals, Medicaid prices are observed and taken as exogenous. For a hospital without an OB unit, I assume its Medicaid price if it did operate one would be the average observed Medicaid price.

To infer Medicaid marginal costs, I assume hospital j's Medicaid marginal cost is

$$\text{MC}_{\text{Medicaid},j} = \boldsymbol{\psi}' \mathbf{w}_{\text{ARHOME},j} - \psi^{\text{Medicaid}} \tag{23}$$

In other words, a Medicaid birth's marginal cost at hospital j is the same as that of the average ARHOME birth at hospital j, minus a Medicaid-specific cost reduction. ARHOME is a natural benchmark because ARHOME plans are Medicaid-funded and insure those near the Medicaid

Table 4: Commercial Price-Setting Results

	Marginal Cost ($1k)
Constant	5.355
	(1.523)
UAMS	0.921
	(0.426)
NICU	0.115
	(0.592)
2×Non-OB Beds	−0.261
	(0.452)
For-Profit	−1.168
	(1.492)
Wage Index (1 SD)	1.775
	(1.130)
ARHOME	−1.267
	(0.414)
Insurers' Relative Bargaining Weight	0.613
	(0.169)
SD of Marginal Cost Shock	1.976
	(0.329)
Observations	1376

Sample: in-network commercial insurer-OB hospital pairs. Pairs where insurer's gain from network inclusion is not positive (8 pairs) and pairs with no births with claims data (10 pairs) are dropped. Marginal cost per unit of intensity. SEs clustered by insurer and system, taking demand estimates as given.

income threshold. A plausible explanation for the cost reduction is supplemental payments, annual lump-sum payments that hospitals receive for treating Medicaid patients. In Arkansas, these payments reimburse hospitals based on how much more revenue they would have earned with Medicare, instead of Medicaid, prices (ACHI, 2023). Thus, a hospital internalizes a lower marginal cost for a Medicaid birth than for an otherwise identical non-Medicaid birth.

Using supplemental payments data, I compute supplemental payments per Medicaid inpatient day among all Arkansas hospitals in 2022. Then, I multiply it by the average number of inpatient days per birth in the sample. I define the result as ψ^{Medicaid}.

The resulting estimate implies a Medicaid birth's marginal cost is $2,238 lower than that of the average ARHOME birth at the same hospital. Implied Medicaid markups are large: among non-outside-option Medicaid births, the average marginal cost is $1,215, and the average markup is $1,782.

Outside Option Prices and Marginal Costs I also require estimates of the outside option's prices and marginal costs. For insurer m, I assume the outside option's price and marginal cost are equal to the average price and marginal cost across insurer m's in-network insurer-OB hospital pairs. The assumed values stay fixed in all counterfactuals.

5 Effects of OB Provision

In this section, I apply the estimated model to answer the following questions. How does OB provision affect access to care, the variable costs of delivering OB care, and maternal and infant health? How does the social value of these effects compare with the private surplus earned by OB hospitals? What are the net externalities of OB provision? I focus on the 37 OB hospitals and quantify the effects of preventing their OB units from closing.

I conduct counterfactual simulations with the following procedure. As a baseline, I simulate equilibrium outcomes under the status quo configuration of OB provision, using realized values of demand-based quality, health-based quality, and marginal cost shocks. Then, for each OB hospital j, I simulate the same equilibrium outcomes if hospital j did not operate an OB unit, holding fixed the OB provision of all other hospitals. The difference between those outcomes and baseline outcomes is the effect of hospital j's OB provision.

First, I measure how each hospital's OB provision affects its private surplus, or the expected variable profit that it generates for the system that owns it. Second, I measure the variable social surplus that it generates through the access-to-care, variable cost, and maternal and infant health effects. I compute the following changes: the change in mothers' demand-based welfare before out-of-pocket spending (i.e., setting out-of-pocket spending to zero), the change in expected variable costs, and the change in expected births with no adverse outcomes.

I convert health effects to monetary units using the social cost of a preterm birth estimated by Waitzman et al. (2021). In doing so, I assume the marginal counterfactual-induced adverse-outcome birth is similar to a preterm birth.[11] Inflation-adjusting to 2022 dollars, the marginal social cost of an adverse-outcome birth is $72,845.

Figure 4 compares an OB hospital's effect on private surplus with each of its three effects on social surplus. The horizontal axis describes an OB hospital's effect on its system's variable profit. The first plot compares it with the access-to-care effect, the second plot compares it with the variable cost effect, and the third plot compares it with the maternal and infant health effect. Each measure of social surplus is divided by the number of Arkansas births.

[11] My measure of health focuses on the extensive margin (adverse-outcome births becoming no-adverse-outcome births, and vice-versa), assuming there are no associated changes on the intensive margin (adverse-outcome births becoming more or less severe). In Appendix B.6, I present descriptive evidence suggesting that extensive-margin

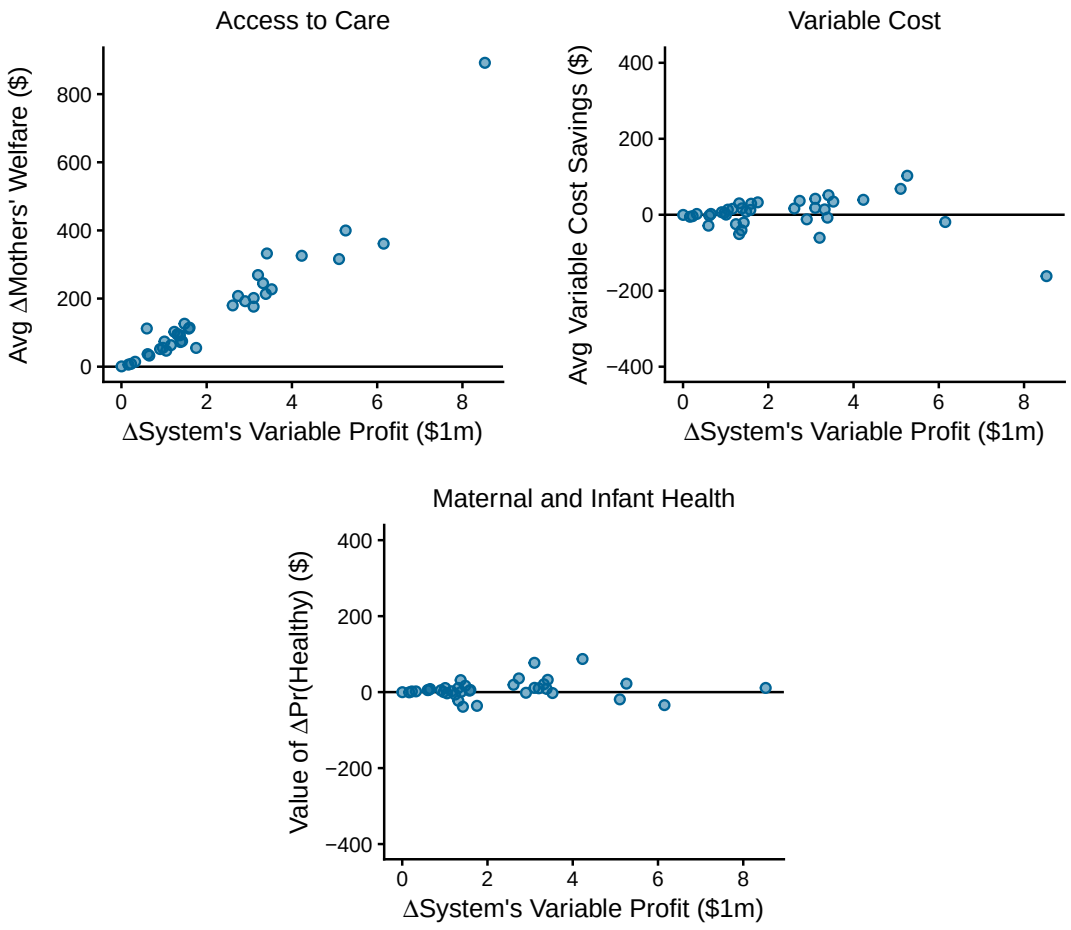


Figure 4: Access-to-Care, Variable Cost, and Maternal and Infant Health Effects

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axes are components of variable social surplus, divided by births: change in mothers' demand-based welfare before OOP, expected variable cost savings, and value of change in expected no-adverse-outcome births caused by hospital's OB provision.

On the one hand, access-to-care effects are positive and large (on average, $162 per Arkansas birth). They are larger for OB hospitals that attract more births and earn higher variable profits. These effects reflect the high value that mothers place on access to more options and shorter travel distances. On the other hand, variable cost effects (on average, $4 per Arkansas birth) and maternal and infant health effects (on average, $8 per Arkansas birth) are small in magnitude and mixed in sign. These effects are driven by the redirection of mothers between high-cost and low-cost, or high-quality and low-quality, hospitals. Some hospitals reduce variable costs or improve health by operating OB units, while others increase variable costs or worsen health by doing so. Given the sizes of these effects, access to care dominates: it accounts for 93% of the average sum of all three components.

health changes are unrelated to intensive-margin health changes.

The sum of all three components, aggregated over Arkansas births, is the total effect on variable social surplus. Figure 5 shows the relationship between an OB hospital's total effect on social surplus and its effect on private surplus. In Section 3, I show that the difference between them is exactly the net externality of OB provision. In the figure, net externalities are depicted as distances between an OB hospital's effect on variable social surplus, plotted on the vertical axis, and its effect on variable profit, denoted by the 45° line.

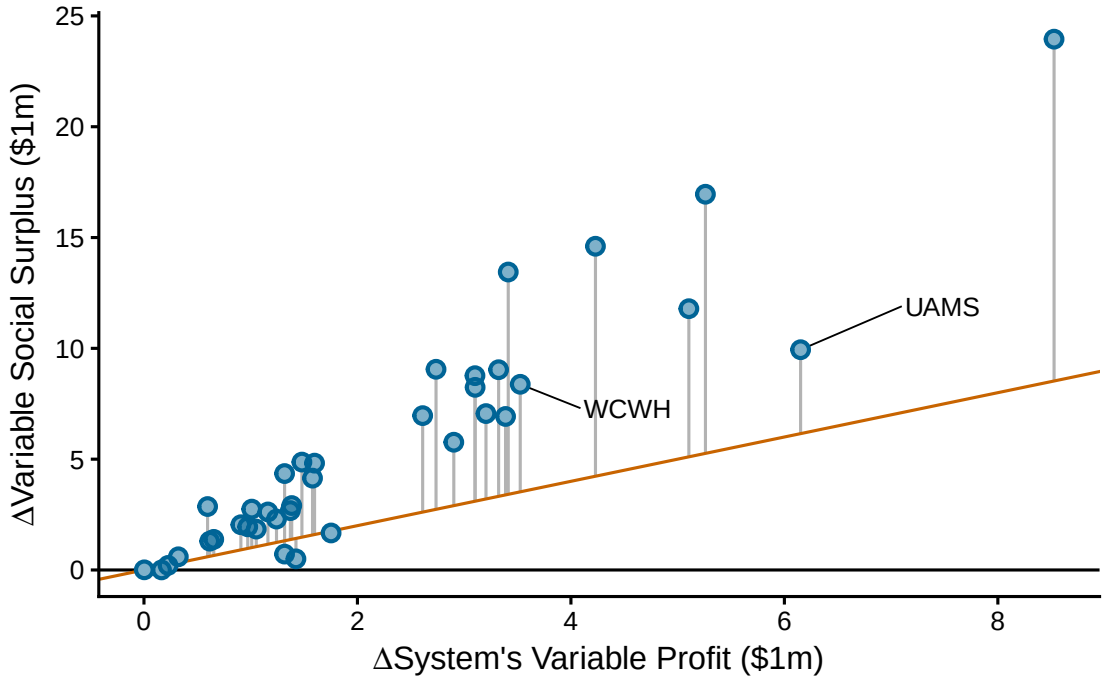


Figure 5: Net Externality of OB Provision

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axis is variable social surplus, summing the following: change in mothers' demand-based welfare before OOP, expected variable cost savings, and value of change in expected no-adverse-outcome births caused by hospital's OB provision. Sloped line is 45° line, and vertical lines denote net externalities. Labeled hospitals are UAMS (teaching) and WCWH (women's).

Almost all OB hospitals have positive externalities, and hospitals that attract more births have larger externalities. On average, an OB hospital earns $2.24 million in variable profit for its system and generates an additional externality of $3.37 million (or $104 per Arkansas birth). As suggested by the comparison in Figure 4, these externalities are mostly driven by the access-to-care effect.[12] Mothers are willing to pay a lot for access to OB units, but OB hospitals do not internalize much of this value. This suggests it is possible for an unregulated market to have too little OB provision relative to the welfare-maximizing outcome.

[12]The access-to-care effect rests on mothers' revealed preferences, so it may be overstated if there are choice frictions. In Appendix G, I show that the average net externality remains positive even with reasonable discounting of mothers' demand-based welfare, including removing the distance-related part entirely.

Additional Results I discuss additional results, with associated figures shown in Appendix D. Figure D1 shows heterogeneity in access-to-care and maternal and infant health effects. Between Medicaid and commercial mothers, and between older and younger mothers, differences in effects are small: each OB hospital equally benefits both groups. However, between rural and urban mothers, each OB hospital tends to mostly benefit one group or the other. On average, OB provision benefits rural mothers more than urban mothers.

Figure D2 shows the distance effects of OB provision: most OB hospitals shorten travel distances, and rural mothers are more affected. Figure D3 shows the role of distance in the access-to-care and maternal and infant health effects. For access to care, distance accounts for a substantial share of total effects, whereas for health, distance effects are near zero while total effects are larger in both positive and negative directions. Figure D4 compares the net externalities of rural and urban OB hospitals, which are broadly similar.

6 Model of OB Provision Decisions

In this section, I extend the empirical model of the hospital OB care market developed in Section 4 to include hospitals' OB provision decisions. Then, I use the model to identify and estimate the fixed cost of OB provision.

I add a Stage 0 that occurs before the two stages discussed in Section 4. In Stage 0, fixed cost shocks are realized. Then, hospital systems make OB provision decisions. Hospital systems have expectations about OB hospital demand, marginal costs, and commercial prices, where uncertainty is over unrealized preference shocks $\boldsymbol{\nu}_i$ for all mothers i, demand-based quality shocks $\boldsymbol{\xi}$, and marginal cost shocks $\mathbf{e}$.

The set of hospitals is taken as exogenous.[13] For hospital j, let M_j denote the set of insurers, including Medicaid, that have it in their networks. Let $\text{Prov}_j \in \{0,1\}$ denote hospital j's OB provision decision. Given all hospitals' OB provision decisions, system s's expected variable profit is

$$\pi_s\left(\mathbf{Prov}\right) = \mathrm{E}\left[\sum_{j \in J_s^{\text{OB}}} \sum_{m \in M_j} \sum_{i \in I_m} \left[s_{ij}\left(J_m^{\text{OB}}, \mathbf{p}_m\right) r_i \left(p_{mj} - \text{MC}_{mj}\right)\right]\right] \tag{24}$$

For hospital j, given all other hospitals' decisions, let $\Delta_j \pi_{s(j)}\left(\mathbf{Prov}_{-j}\right)$ denote the change in its system's expected variable profit if it switches from non-provision to provision. I assume systems play a pure-strategy subgame perfect equilibrium. This rules out profitable single-hospital deviations: given all other hospitals' decisions, hospital j operates an OB unit if the

[13] Hospital openings and closures are rarer than OB unit closures, and they are unlikely to be driven by OB care. OB care accounts for 3.2% of revenue at OB hospitals.

effect of its OB provision on its system's expected total profit is positive, and it does not if the effect is negative. In other words,

$$\text{Prov}_j = \begin{cases} 1 & \text{if } \Delta_j \pi_{s(j)} \left(\mathbf{Prov}_{-j} \right) > \text{FC}_j \\ 0 & \text{if } \Delta_j \pi_{s(j)} \left(\mathbf{Prov}_{-j} \right) < \text{FC}_j \end{cases} \tag{25}$$

where FC_j is fixed cost.

I further parameterize fixed cost as

$$\text{FC}_j = \overline{\text{FC}} + u_j, \quad \text{E}\left[u_j\right] = 0 \tag{26}$$

where $\overline{\text{FC}}$ is expected fixed cost and u_j is an unobserved fixed cost shock.

Discussion of OB Provision Model I highlight four notes about the model of OB provision. First, the OB provision decision is modeled as static, so it does not depend on past or future outcomes. In Appendix E, I discuss reasonable assumptions under which a static model captures all relevant features of OB provision.

Second, I follow much of the literature on fixed cost estimation in assuming demand-based quality and marginal cost shocks are not realized until after firms decide whether to participate in the market. Thus, participants and non-participants have the same distribution of those shocks.[14] To support this assumption, the utility and marginal cost equations (Equations 9 and 21) include the most salient and relevant hospital characteristics, so it is plausible that any remaining variation is unanticipated.

Third, Equation 25 only rules out profitable single-hospital deviations. An equilibrium also rules out profitable multi-hospital deviations, such as a system switching out one OB hospital for another. Empirically, I find that ruling out multi-hospital deviations does not provide any additional information in this setting.

Fourth, just like marginal cost, fixed cost reflects not just direct accounting costs (e.g., attending physicians, equipment, or liability insurance), but also indirect costs and benefits from operating an OB unit for a year, such as reputational benefits.

Identification of Expected Fixed Cost The parameter to identify is $\overline{\text{FC}}$. For all mothers i and hospitals j, I observe r_i and Prov_j. Also, the distributions of $s_{ij}(\cdot)$, $\text{MC}_{m(i)j}$, and $p_{m(i)j}$ are identified from the OB hospital demand and commercial price-setting stages.

[14] An alternative approach is to relax this assumption and estimate all stages of the model jointly to get estimates of those different distributions (Ciliberto et al., 2021). In this paper, this is infeasible given the complexity of the OB hospital demand and commercial price-setting stages.

Following Eizenberg (2014), I assume the support of fixed cost is in the support of the effect each hospital's OB provision has on its system's expected variable profit, given the observed OB provision of all other hospitals. In other words,

$$\text{FC}_j \in \left[\inf_{j' \in J} \Delta_{j'} \pi_{s(j')} \left(\mathbf{Prov}_{-j'} \right), \sup_{j' \in J} \Delta_{j'} \pi_{s(j')} \left(\mathbf{Prov}_{-j'} \right) \right] \quad \text{for all hospitals } j \tag{27}$$

This assumption is equivalent to the following statement: given the observed OB provision of all other hospitals, the most profitable hospital would operate an OB unit for any possible fixed cost shock, and the least profitable hospital would never operate an OB unit for any possible fixed cost shock.

This assumption, along with Equation 25, implies each FC_j has lower and upper bounds

$$L_j = \begin{cases} \inf_{j' \in J} \Delta_{j'} \pi_{s(j')} \left(\mathbf{Prov}_{-j'} \right) & \text{if } \text{Prov}_j = 1 \\ \Delta_j \pi_{s(j)} \left(\mathbf{Prov}_{-j} \right) & \text{if } \text{Prov}_j = 0 \end{cases} \tag{28}$$

$$U_j = \begin{cases} \Delta_j \pi_{s(j)} \left(\mathbf{Prov}_{-j} \right) & \text{if } \text{Prov}_j = 1 \\ \sup_{j' \in J} \Delta_{j'} \pi_{s(j')} \left(\mathbf{Prov}_{-j'} \right) & \text{if } \text{Prov}_j = 0 \end{cases} \tag{29}$$

Let $\tilde{\mathbf{y}}_j$ be instruments plausibly assumed to be related to OB provision but independent of fixed cost shock u_j. Then, Equation 26 implies

$$\overline{\text{FC}} \geq \text{E}\left[L_j \mid \tilde{\mathbf{y}}_j \right] \tag{30}$$

$$\overline{\text{FC}} \leq \text{E}\left[U_j \mid \tilde{\mathbf{y}}_j \right] \tag{31}$$

The identified set of $\overline{\text{FC}}$ is the set of values that satisfy the moment inequalities in Equations 30 and 31 for all values of $\tilde{\mathbf{y}}_j$. I include one instrument in $\tilde{\mathbf{y}}_j$: expected total intensity-weighted quantity if all hospitals operated OB units and all prices were equal to the average observed price, discretized into equal-sized bins.

For intuition, $\overline{\text{FC}}$ is identified by the relationship between OB provision and profitability: if expected fixed cost is higher, then hospitals need to earn higher variable profits to justify operating OB units. Conditioning on the instrument sharpens the identified set by forcing the estimate to explain differences in OB provision between attractive and unattractive hospitals under plausible restrictions on fixed cost shocks.

Estimation of Expected Fixed Cost I compute empirical analogs of the worst-case fixed cost bounds in Equation 27. I correct for the upward bias of a sample minimum and downward

bias of a sample maximum following Hall and Park (2002).

Using empirical analogs of the moment inequalities in Equations 30 and 31, I estimate expected fixed cost. An uncorrected estimated set would be the range between the highest average lower bound and the lowest average upper bound over all instrument bins. However, due to estimation error, the highest average is biased upward, and the lowest average is biased downward. I apply the estimation and inference procedure of Chernozhukov et al. (2013), which corrects each average according to its imprecision. Confidence intervals are clustered by system, and they take demand and commercial price-setting estimates as given.

Table 5 shows the results. The main specification discretizes the quantity instrument into 10 equal-sized bins. As a robustness check, I show how results vary for different numbers of bins. The main specification implies expected fixed cost is between $0.35 and 2.21 million. This result relies on the instrument, but it is not very sensitive to reasonable changes in the number of instrument bins.

Table 5: OB Provision Results

	Expected Fixed Cost ($1m)	
Instrument Bins	Estimated Set	95% CI
1 (No Instrument)	[0.290, 9.334]	[0.181, 11.122]
8	[0.336, 3.624]	[0.057, 4.390]
9	[0.355, 3.330]	[0.115, 4.023]
10 (Main)	[0.346, 2.214]	[0.001, 2.602]
11	[0.401, 2.818]	[−0.071, 3.366]
12	[0.234, 4.305]	[−0.011, 5.520]
Observations	71	

Sample: all hospitals. Main specification is 10 bins. Bias-corrected estimates and CIs following Chernozhukov et al. (2013). System-clustered CIs, taking demand and commercial price-setting estimates as given.

7 Policies to Prevent OB Unit Closures

In this section, I apply the extended model to evaluate two policies, Arkansas's 2025 Healthy Moms, Healthy Babies Act and an OB unit subsidy. These policies are aimed at preventing OB unit closures, which are especially salient in Arkansas. In Appendix B.1, I show the history of OB provision in this setting: there were 24 closures of OB units and zero openings of OB units in existing hospitals between 1995 and 2024.

Many commonly reported reasons for OB unit closures are related to fixed costs, such as the costs of retaining attending physicians and liability insurance (Busse et al., 2025). As realistic

baseline scenarios, I simulate three conditions that may induce OB unit closures: \$100,000, \$200,000, and \$300,000 increases in all fixed costs.

In all counterfactuals, I focus on OB unit closures and not openings. I allow the 37 OB hospitals to change their OB provision decisions, but I assume the 34 non-OB hospitals never operate OB units. This is consistent with historical trends, and it can be explained by one-time sunk costs of building a brand-new OB unit.

I make the following assumptions about unobserved shocks. Hospital systems make OB provision decisions taking expectations over unrealized demand-based quality and marginal cost shocks. However, when computing outcomes, I assume demand-based quality, health-based quality, marginal cost, and fixed cost shocks always take on the same realized values as the status quo. The first three shocks are estimated; fixed cost shocks are not, so I assume a uniform prior distribution and take draws from the posterior distribution induced by observed OB provision.

Specifically, I use the following procedure. First, I choose a value of expected fixed cost $\overline{\text{FC}}$ from the estimated set. Then, I compute the set of all possible fixed cost shock vectors $\mathbf{u}$ that satisfy two conditions: observed OB provision is rationalized (i.e., $\text{FC}_j \in \left[L_j, U_j\right]$ for all hospitals j), and instrument exogeneity is empirically satisfied (i.e., $\text{Avg}\left(u_j \mid \tilde{\mathbf{y}}_j\right) = 0$ for all values of $\tilde{\mathbf{y}}_j$). I take draws of $\mathbf{u}$ from a uniform distribution over this set. For each draw, I solve for an equilibrium configuration of OB provision and simulate resulting outcomes. I take an average over draws of $\mathbf{u}$: this is the predicted outcome. Finally, I repeat this procedure for all values of $\overline{\text{FC}}$ in the estimated set and report the range of predicted outcomes.

In the step that requires solving for an equilibrium configuration of OB provision, I use a sequential algorithm. I start from the status quo configuration, and I compute each system's total-profit-maximizing configuration for its own hospitals, given the OB provision of all other hospitals. Then, I allow the system with the highest gain from deviating to change its configuration. I repeat this process, allowing one system to change its configuration at a time, until there are no more profitable deviations.[15]

To benchmark this exercise, I examine actual OB unit closures in the two years after the sample period but before the policy (2023–2024). In that time, 4 hospitals closed their OB units.[16] Table 6 shows the following predictions for each simulated scenario without any policy: predicted OB unit closures, the share of actual-closers predicted to close, and the share of other OB units predicted to close.

[15]Two issues may arise with this approach. First, the algorithm may not converge, possibly due to equilibrium nonexistence. Empirically, I find convergence in 99.8% of simulations. Second, the algorithm selects one equilibrium when multiple equilibria may exist. I find that alternative algorithms do often select different equilibrium configurations, but reported results, such as those in Tables 7 and 8, do not change much.

[16]Data on actual OB unit closures come from UAMS's Arkansas Maternal Health Scorecard.

Table 6: Predicted OB Unit Closures

	OB Closures (Out of 37)	OB Closures (%)	
		Hospitals With Actual Closures in 2023–2024 (Out of 4)	Others (Out of 33)
\$100k Fixed Cost Increase	[2.1, 6.2]	[31.4, 36.0]	[2.2, 14.9]
\$200k Fixed Cost Increase	[3.3, 7.8]	[41.6, 46.7]	[4.6, 18.7]
\$300k Fixed Cost Increase	[4.5, 9.7]	[54.1, 57.4]	[6.7, 22.7]

Source for actual OB unit closures: UAMS. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost.

First, the actual number of OB unit closures, four, lies in the middle of the range of predictions under a \$100,000 fixed cost increase. Thus, I interpret the \$100,000 increase as a reasonable approximation of market conditions in Arkansas in 2023–2024. I interpret the \$200,000 increase as a more pessimistic approximation of the same period or the years just after, and the \$300,000 increase as conditions further into the future.

Second, I compare the 4 hospitals that actually closed their OB units with the 33 that did not. Actual-closers have much higher predicted chances of closing: after a \$100,000 fixed cost increase, an actual-closer closes in 31% to 36% of simulations, in contrast to 2% to 15% for other OB hospitals. This is an out-of-sample validation of model fit because I do not use data on actual OB unit closures for estimation.

Healthy Moms, Healthy Babies Act The first policy I evaluate is the Medicaid price hike from the Healthy Moms, Healthy Babies Act. I consider the actual hike of 70% as well as a smaller hike of 35%. One outcome of interest is total social welfare: mothers' demand-based welfare before out-of-pocket spending, plus the value of expected births with no adverse outcomes, minus expected variable costs, minus fixed costs. I also compute the marginal value of public funds (Hendren & Sprung-Keyser, 2020), or the change in private welfare per dollar of public spending, where public spending is all spending by Medicaid. Just like in Section 5, I convert health effects to monetary units by assuming the marginal social cost of an adverse-outcome birth is \$72,845, based on the estimated social cost of a preterm birth from Waitzman et al. (2021). Table 7 shows the results.

The 70% Medicaid price hike prevents many OB unit closures. With the policy in place, only 1 OB unit closes after a \$100,000 fixed cost increase, around 2 close after a \$200,000 increase, and around 3 to 4 close after a \$300,000 increase. The policy improves social welfare by \$1 to 12 million after a \$100,000 increase, \$2 to 13 million after a \$200,000 increase, and \$3 to 16 million after a \$300,000 increase. The policy costs \$29.0 to 29.2 million in public spending, which is

Table 7: Evaluation of the Healthy Moms, Healthy Babies Act

	OB Closures (Out of 37)	ΔSocial Welfare ($1m)	ΔPublic Spending ($1m)	MVPF
$100k Fixed Cost Increase	[2.1, 6.2]	–	–	–
+ Medicaid Price +35%	[1.3, 1.5]	[+0.8, +11.5]	[+14.5, +14.6]	[1.06, 1.79]
+ Medicaid Price +70%	1	[+0.8, +11.6]	[+29.0, +29.1]	[1.03, 1.40]
$200k Fixed Cost Increase	[3.3, 7.8]	–	–	–
+ Medicaid Price +35%	[2.3, 5.6]	[+1.4, +8.5]	[+14.5, +14.6]	[1.09, 1.58]
+ Medicaid Price +70%	[1.7, 1.9]	[+1.7, +13.3]	[+29.0, +29.1]	[1.06, 1.46]
$300k Fixed Cost Increase	[4.5, 9.7]	–	–	–
+ Medicaid Price +35%	[3.4, 7.5]	[+1.9, +7.5]	+14.5	[1.13, 1.51]
+ Medicaid Price +70%	[2.6, 3.7]	[+2.7, +15.7]	[+29.0, +29.2]	[1.09, 1.54]

Actual policy is 70% Medicaid price hike. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Welfare consists of mothers' demand-based welfare before OOP, plus value of expected no-adverse-outcome births, minus expected variable cost, minus fixed cost. MVPF=1+ΔWelfare/ΔPublic Spending.

similar to the policy's stated annual budget of $26 million (Arkansas Governor, 2025).

Across all fixed cost scenarios and all values of expected fixed cost in the estimated set, the policy's marginal value of public funds ranges from 1.0 to 1.5. In other words, each dollar of spending is a dollar transferred to hospitals plus $0 to $0.50 of additional social welfare.

In the case of a $100,000 fixed cost increase, a 35% Medicaid price hike obtains nearly all of the welfare gain of the full 70% hike at half the cost, so its marginal value of public funds is higher. However, for larger fixed cost increases, it obtains less of the full hike's gains. A smaller policy may be more cost-effective than the full policy in the short term, but its advantage is less pronounced further into the future.

OB Unit Subsidy One drawback of a Medicaid price hike is that it pays OB hospitals in proportion to their Medicaid births. Thus, it pays larger amounts to OB hospitals that attract more births, earn higher variable profits, and would have operated OB units anyway. Meanwhile, the OB units most at risk of closing receive less; even with a 70% Medicaid price hike, 1 to 4 OB units still close.

As an alternative, I consider a flat OB unit subsidy: an annual payment of the same amount to every hospital that operates an OB unit. I consider subsidies of $100,000, $200,000, and $300,000, up to the size of the fixed cost increase. In this exercise, public spending includes both Medicaid spending and the cost of the subsidy. Table 8 shows the results.

A subsidy equal to the fixed cost increase keeps all OB units open at a fraction of the cost of the Healthy Moms, Healthy Babies Act. In contrast to the Medicaid price hike's $29 million price

Table 8: Evaluation of OB Unit Subsidy

	OB Closures (Out of 37)	ΔSocial Welfare ($1m)	ΔPublic Spending ($1m)	MVPF
$100k Fixed Cost Increase	[2.1, 6.2]	–	–	–
+ OB Subsidy $100k	0	[+0.9, +11.6]	+3.7	[1.24, 4.17]
$200k Fixed Cost Increase	[3.3, 7.8]	–	–	–
+ OB Subsidy $100k	[2.1, 6.2]	[+0.9, +4.2]	[+3.1, +3.5]	[1.26, 2.25]
+ OB Subsidy $200k	0	[+1.8, +13.5]	[+7.4, +7.5]	[1.24, 2.83]
$300k Fixed Cost Increase	[4.5, 9.7]	–	–	–
+ OB Subsidy $100k	[3.3, 7.8]	[+1.3, +5.5]	[+3.0, +3.4]	[1.39, 2.84]
+ OB Subsidy $200k	[2.1, 6.2]	[+2.2, +7.8]	[+6.3, +7.0]	[1.31, 2.18]
+ OB Subsidy $300k	0	[+3.1, +19.1]	[+11.1, +11.2]	[1.28, 2.70]

Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Welfare consists of mothers' demand-based welfare before OOP, plus value of expected no-adverse-outcome births, minus expected variable cost, minus fixed cost. MVPF=1+ΔWelfare/ΔPublic Spending.

tag, a fully offsetting subsidy only increases public spending by $4 million after a $100,000 fixed cost increase, $7 million after a $200,000 increase, and $11 million after a $300,000 increase. It improves social welfare by $1 to 12 million, $2 to 14 million, and $3 to 19 million, respectively, which is at least as much as the Medicaid price hike in every scenario and at every value of expected fixed cost in the estimated set. As a result, its marginal value of public funds is higher, ranging from 1.2 to 4.2.

Smaller subsidies are also more cost-effective than the Medicaid price hike. A $100,000 subsidy after a $200,000 or $300,000 fixed cost increase, and a $200,000 subsidy after a $300,000 fixed cost increase, generate less welfare than the Medicaid price hike but at much lower cost. Their marginal values of public funds range from 1.3 to 2.8.

As a caveat, there are reasons outside the scope of this paper to prefer a Medicaid price hike over an OB unit subsidy. One is financing: the federal government pays a large fraction of the cost of state Medicaid programs. Thus, a Medicaid price hike is cheaper to the Arkansas state government than the public spending numbers imply. Another is incentives: a higher Medicaid price rewards hospitals for treating Medicaid patients, which may alleviate administrative burdens or improve Medicaid patients' quality of care. Finally, a Medicaid price hike uses existing payment systems, but a subsidy would require developing a new program and setting guidelines about what qualifies as an OB unit.

Additional Results I discuss additional results, with the associated tables shown in Appendix F. Table F1 decomposes the policies' effect on social welfare into its components. For both policies, almost all of the gain is access to care. Changes in maternal and infant health and

changes in variable costs are small and mixed in sign. Much of the gain, however, is offset by fixed costs because more OB units are kept open.

Tables F2, F3, and F4 show heterogeneity in the policies' access-to-care and maternal and infant health effects. Both policies have similar effects on Medicaid and commercial mothers, as well as on older and younger mothers. The largest difference is between rural and urban mothers: rural mothers benefit more.

Table F5 shows the policies' distance effects. Both policies shorten travel distances, with larger reductions for rural mothers. Table F6 shows the role of distance in the policies' access-to-care and maternal and infant health effects. Distance has more impact on access to care than on health. Table F7 shows OB unit closures among rural and urban hospitals, which both policies affect roughly equally.

Finally, in Appendix G, I show that the policy conclusions in this section are robust to reasonable discounting of mothers' demand-based welfare to account for the possibility of choice frictions.

8 Conclusion

I estimate a model of the hospital OB care market in Arkansas to answer the following questions. How does OB provision affect access to care, the costs of delivering care, and maternal and infant health in equilibrium? What are the net externalities of OB provision? What are the consequences of policies aimed at preventing OB unit closures?

I find that OB provision benefits mothers substantially through access to care. This is driven by the high value that mothers place on access to more options and shorter travel distances. In contrast, variable cost effects and maternal and infant health effects are small and mixed in sign. Overall, OB hospitals generate positive net externalities, on average worth $104 per Arkansas birth.

This gap between social and private surplus suggests a role for policy. Under realistic increases in fixed costs, the Healthy Moms, Healthy Babies Act's 70% Medicaid price hike prevents many closures. Its marginal value of public funds is 1.0 to 1.5, and its benefits come primarily through access to care. However, a Medicaid price hike is a costly way to prevent OB unit closures because it pays larger amounts to hospitals that would have operated OB units anyway. Alternatively, a flat subsidy to all OB hospitals that fully offsets the fixed cost increase prevents every closure and generates at least as much social welfare as the Healthy Moms, Healthy Babies Act. Its marginal value of public funds is 1.2 to 4.2. This suggests there are notable gains to policies that better target OB units most at risk of closing.

There are several promising areas for future research related to this paper. First, I abstract

away from capacity constraints (Avdic et al., 2024) and effects of patient volumes on quality or efficiency (Avdic et al., 2019; Dingel et al., 2026; Domenella, 2026). These frictions create additional consequences to reallocating patients, potentially amplifying the effects of OB provision. Second, I do not consider spillover effects, which may be important; for example, one hospital's OB provision may make inputs scarcer for other OB hospitals (Cutler et al., 2010). Third, I assume hospitals cannot influence their own demand-based quality, health-based quality, or marginal costs. Future research, by modeling OB hospitals' decisions to invest in quality or efficiency, can evaluate policies that encourage OB hospitals to become higher-quality or lower-cost (Garthwaite et al., 2022; Hackmann, 2019).

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A Data: Additional Details

A.1 Main Data Construction

I only include mothers who live in the setting: mothers who live outside of Arkansas or in metropolitan areas with population centers in other states (Crittenden, Miller, or Little River Counties) are dropped. For multiple births (e.g., twins or triplets), I drop births after the first-occurring one. Births that occur outside of in-network OB hospitals in the setting (e.g., out-of-network OB hospitals, OB hospitals in dropped counties, non-OB hospitals, birth centers, or home births) are assigned to the outside option.

I observe mothers' home locations at the county level. I measure the haversine distances

between the centroids of census block groups and hospitals, construct each mother's probability of living in each census block group using her county and age, and impute mothers' expected distances to hospitals.

I assign each birth to claims data using the following procedure. First, I match on a person identifier (encrypted combination of mother's name, mother's date of birth, and mother's sex). Then, I filter matches to those where the claims' dates of insurance coverage and dates of service overlap with the date of birth. Then, I filter claims to those with diagnosis or procedure codes related to labor and delivery. Finally, I filter matches to those where the claims' National Provider Identifiers (NPIs) are linked to the birth's OB hospital, either by geographic proximity or by frequency of matching claims.

There are two reasons why births may not match to claims data. First, the claims data are not comprehensive. Mothers may be missing if they pay entirely out-of-pocket, are uninsured, or are insured by entities not required to report to the All-Payer Claims Database (e.g., employer-sponsored self-insured plans, plans with fewer than 2,000 enrollees, Medicare, or Veterans Affairs). Second, the match between datasets may not be perfect due to ambiguity in mothers' names or other data collection errors.

I define an insurer as a combination of payer type (e.g., Medicaid, ARHOME, or employer-sponsored), submitter (e.g., Blue Cross Blue Shield), group (e.g., employer), and plan. Among payer-submitter-group-plan combinations with fewer than 10 births with claims data, I aggregate them into one insurer per payer-submitter-group. Among payer-submitter-group combinations with fewer than 10 births with claims data, I aggregate them into one insurer per payer-submitter. Among payer-submitter combinations with fewer than 10 births with claims data, I aggregate them into one insurer per payer. Finally, if any insurers still have fewer than 10 births with claims data, I drop them and their associated births from the sample.

I define hospital system s as in-network for insurer m if either of the following is true: there is at least one birth with claims data at system s from insurer m's enrollees, or there are at least 10 non-birth-related claims (which I match to hospitals using NPIs and geographic proximity) at system s from insurer m's enrollees. A system being in-network is equivalent to all of its hospitals being in-network. All hospitals are in-network for Medicaid.

Finally, for insurers with fewer than 2 in-network OB hospitals, I drop them and their associated births from the sample.

A.2 Recovering Intensity, Price, and Out-of-Pocket Spending

For mother i with insurer m, I observe her payment and out-of-pocket spending (split among copay, deductible, and coinsurance). I assume payment and out-of-pocket spending are

$$\text{Payment}_{ij} = r_i p_{mj} \tag{A1}$$

$$\text{OOP}_{ij} = \text{Copay}_i + \text{Deductible}_i + \text{CoinsuranceRate}_i \left(\text{Payment}_{ij} - \text{Copay}_i - \text{Deductible}_i \right) \tag{A2}$$

where r_i is resource use intensity and p_{mj} is price per unit of intensity.

To recover r_i and p_{mj}, I estimate

$$\ln \text{Payment}_{ij} = \boldsymbol{\rho}' \mathbf{x}_i + \text{FE}_{mj} + \upsilon_{ij}, \quad \text{E}\left[\upsilon_{ij}\right] = 0 \tag{A3}$$

where $\mathbf{x}_i$ consist of a constant, mother's age, mother's race, mother's BMI, a diabetes indicator, a hypertension indicator, a smoker indicator, a prior birth indicator, a prior caesarean section indicator, logged home county population density, and home county population shares by income bin. For intuition, the estimate of r_i is approximately $\exp\left(\hat{\boldsymbol{\rho}}' \mathbf{x}_i\right)$, and the estimate of p_{mj} is approximately $\exp\left(\widehat{\text{FE}}_{mj}\right)$.

To turn those approximations into unbiased estimates, I make two adjustments. The first adjustment is to apply empirical Bayes shrinkage to $\widehat{\text{FE}}_{mj}$. To do so, I estimate

$$\text{FE}_{mj} = \text{FE}_m + \text{FE}_j + \tilde{\upsilon}_{mj}, \quad \text{E}\left[\tilde{\upsilon}_{mj}\right] = 0 \tag{A4}$$

where FE_{mj}, FE_m, and FE_j are estimated with normal error. I obtain shrunk versions of $\widehat{\text{FE}}_m$ and $\widehat{\text{FE}}_j$. Then, $\widehat{\text{FE}}_{mj}$ is shrunk toward $\widehat{\text{FE}}_m + \widehat{\text{FE}}_j$. If insurer-hospital pair mj has no births with claims data, then $\widehat{\text{FE}}_{mj} = \widehat{\text{FE}}_m + \widehat{\text{FE}}_j$ exactly. As a note, the unshrunk prices are maintained in one place: I subtract markups from unshrunk prices to obtain implied marginal costs when estimating commercial price-setting parameters.

The second adjustment is to correct the estimates of r_i and p_{mj} to account for Jensen's inequality because estimation errors in $\hat{\boldsymbol{\rho}}$ and $\widehat{\text{FE}}_{mj}$ enter non-linearly.

Finally, I multiply r_i by a constant, and divide p_{mj} by the same constant, so the average of r_i is 1. Then, p_{mj} can be interpreted as the price of an average birth.

B Additional Summary Statistics and Descriptives

B.1 History of OB Provision in Arkansas

Figure B1 shows the history of OB provision in this setting in 1995–2024. There are few hospital openings and closures: 6 openings and 7 closures. There are also few positive changes in OB provision: 2 openings of OB hospitals and zero additional OB units in existing hospitals. Most frequent are negative changes in OB provision: 2 closures of OB hospitals and 24 closed OB units in existing hospitals. In such a setting where OB unit closures are frequent and OB unit openings are not, encouraging OB provision is equivalent to preventing OB unit closures.

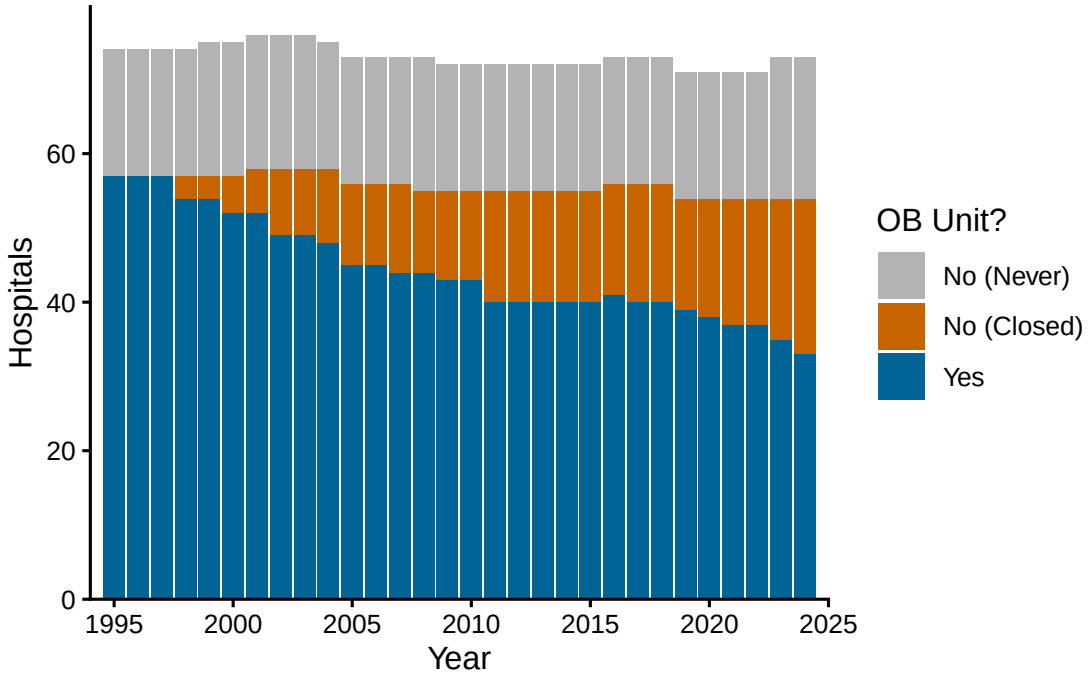


Figure B1: History of OB Provision in Arkansas

Data source: AHA. Sample: hospitals in Arkansas (except Crittenden/Miller/Little River Counties and specialty/VA hospitals).

B.2 Summary Statistics

Table B1 shows birth summary statistics split by whether they match to claims data. Table B2 shows birth summary statistics by quarter of the year. Table B3 shows rates of different adverse outcomes. Table B4 shows hospital-level summary statistics.

Table B1: Birth Summary Statistics, by Match to Claims Data

	Commercial (Matched)	Commercial (Not Matched)	Medicaid (Matched)	Medicaid/Other (Not Matched)
Mean				
Distance (mi)	17.4	15.7	17.1	15.3
Any Adverse Outcome	0.03	0.02	0.03	0.03
Age (yr)	28.0	29.2	25.7	27.4
BMI	28.9	28.6	29.0	28.7
Prior Births>0	0.58	0.56	0.66	0.70
Prior C-sections>0	0.14	0.15	0.16	0.14
Rural	0.34	0.25	0.39	0.31
Prenatal Visits	10.8	11.2	9.3	9.2
Gestational Age (wk)	38.2	38.3	38.1	38.3
Planned C-section	0.22	0.21	0.22	0.20
Outside Option	0.013	0.003	0.011	0.083
UAMS (Teaching)	0.086	0.053	0.117	0.154
WCWH (Women's)	0.048	0.054	0.059	0.107
Public	0.10	0.06	0.15	0.18
Non-profit	0.72	0.76	0.65	0.52
For-profit	0.16	0.18	0.20	0.22
NICU	0.54	0.60	0.50	0.55
Non-OB Beds	296.1	308.3	250.6	230.0
Observations	9561	8442	10526	3778

Adverse outcomes: maternal (perineal laceration, blood transfusion, unplanned hysterectomy, uterine rupture) or infant (Apgar<4, death, injury, seizure).

B.3 Descriptive Evidence About Home Locations

I present descriptive evidence suggesting that conditional on observed characteristics, mothers' home locations are unrelated to OB care, supporting the empirical model's assumption that home locations are exogenous. There are two types of possible violations that would be problematic. First, mothers who are unobservably more sensitive to distance (or unobservably sicker), may tend to live nearer to or farther from OB hospitals, biasing estimates related to distance. Second, mothers who are unobservably more sensitive to any hospital characteristic k (or unobservably sicker) may tend to live nearer to or farther from OB hospitals with characteristic k, biasing estimates related to k.

I examine plausible channels through which home location may relate to unobserved preference or underlying health shocks. As proxies for these channels, I use mothers' characteristics not included in estimation: marital status, high school completion, hepatitis C diagnosis, height, and a measure of environmental exposure, outdoor nitrogen dioxide (NO_2) concentration by census block group.[17] To summarize a mother's home location, I consider the following at-

[17] Data on NO_2 concentration in 2020 come from the Environmental Protection Agency's Environmental Justice Screening and Mapping Tool. I impute each mother's expected value based on her county and age.

Table B2: Birth Summary Statistics, by Quarter

	Jan–Mar	Apr–Jun	Jul–Sep	Oct–Dec
Mean				
Distance (mi)	16.7	16.4	16.9	16.4
Any Adverse Outcome	0.03	0.03	0.03	0.03
Payment ($)	5614	5524	5477	5588
Out-of-pocket ($)	552	509	470	443
Commercial	0.56	0.56	0.55	0.56
Medicaid	0.41	0.41	0.42	0.41
Other	0.04	0.02	0.03	0.03
Age (yr)	27.5	27.5	27.3	27.6
BMI	28.8	28.9	28.7	28.9
Prior Births>0	0.61	0.62	0.61	0.62
Prior C-sections>0	0.14	0.15	0.15	0.15
Rural	0.33	0.32	0.34	0.31
Prenatal Visits	10.1	10.3	10.2	10.4
Gestational Age (wk)	38.2	38.2	38.2	38.2
Planned C-section	0.21	0.22	0.22	0.21
Outside Option	0.017	0.020	0.018	0.018
UAMS (Teaching)	0.096	0.096	0.092	0.099
WCWH (Women's)	0.064	0.061	0.058	0.057
Public	0.12	0.12	0.11	0.12
Non-profit	0.68	0.68	0.69	0.68
For-profit	0.19	0.18	0.18	0.18
NICU	0.54	0.55	0.53	0.55
Non-OB Beds	275.3	277.0	271.6	283.2
Observations	7835	7776	8530	8166

Adverse outcomes: maternal (perineal laceration, blood transfusion, unplanned hysterectomy, uterine rupture) or infant (Apgar<4, death, injury, seizure).

tributes: distance to the nearest OB hospital in the mother's choice set, whether that nearest option is UAMS, whether it has a NICU, and its logged non-OB beds. Using ordinary least squares, I regress each attribute on each proxy, controlling for the mothers' characteristics that appear in either the demand model (Equations 10 and 11) or health model (Equation 12). Then, I report the coefficient on the proxy. Table B5 shows the results.

All coefficients are near zero and statistically insignificant. This rules out the most plausible channels through which mothers' home locations may relate to unobserved shocks, suggesting that any variation in mothers' home locations that remains after conditioning on observed characteristics is driven by factors unrelated to OB care.

Table B3: Adverse Outcome Rates

	Mean
Maternal	
Perineal Laceration	0.0065
Blood Transfusion	0.0055
Unplanned Hysterectomy	0.0005
Uterine Rupture	0.0003
Infant	
Apgar<4	0.0101
Death	0.0068
Injury	0.0005
Seizure	0.0002
Observations	32307

Apgar: overall infant health 5 min after birth, scored 0–10, <4 is critical.

B.4 Descriptive Evidence About Insurer Choice

If mothers strategically choose insurers for reasons related to OB care (e.g., out-of-pocket spending or networks), then the empirical model's assumption that insurers are exogenous would be violated, and the model's estimates would be biased. I present two sets of descriptive evidence suggesting that this is not the case.

First, I examine how many insurers a mother may plausibly be able to choose from. Table B6 shows the number of insurers available to each mother under the following assumptions. First, a mother only chooses among insurers in her own payer type. In other words, a Medicaid mother can only choose Medicaid, an ARHOME mother only chooses among ARHOME plans, a state employee only chooses among state employee plans, and a mother whose plan is from the Affordable Care Act (ACA) marketplace only chooses among plans from the ACA marketplace. Second, a mother with employer-sponsored insurance has an additional restriction: she only chooses among plans in the same group (which typically corresponds to employer).

Medicaid mothers only have one option, and mothers with employer-sponsored and state employee insurance have few options (on average 1.3 and 7, respectively). Together, those mothers account for 64% of the sample. I cannot rule out the possibility that the remainder, ARHOME mothers and mothers who shop on the ACA marketplace, have many options (up to 23 and 19, respectively). However, there are likely additional restrictions, such as plans being region-locked or only offered at certain times of year, that I do not account for.

Second, I show that even if mothers have access to multiple options, they do not strategically choose insurers for reasons related to OB care. For each birth from October 9 to December 31, I examine whether the mother was enrolled with the birth insurer before the start of pregnancy

Table B4: Hospital Summary Statistics

	OB Unit	No OB Unit
Count		
UAMS (Teaching)	1	0
WCWH (Women's)	1	0
Public	4	5
Non-profit	25	26
For-profit	8	3
NICU	9	0
Mean		
Non-OB Beds	157.5	53.0
Patient Reviews (1 SD)	−0.38	0.41
Patient Outcomes (1 SD)	−0.29	0.31
Wage Index (1 SD)	0.3	−0.3
Observations	37	34

Sources: AHA, CMS. Patient reviews: composite of 10 questions in Medicare patient survey, normalized to mean 0 and SD 1. Patient outcomes: composite of Medicare mortality and readmission rates for pneumonia, COPD, and heart failure, normalized to mean 0 and SD 1. Wage index: regional hospital worker wage, normalized to mean 0 and SD 1.

(280 days before birth). I also examine whether, at any point during pregnancy, the mother is enrolled with any insurer other than the birth insurer. Table B7 shows the results.

Most mothers (74% of commercial and 62% of Medicaid mothers) are enrolled with their birth insurers before the start of pregnancy, suggesting that those insurers are not chosen for reasons related to OB care.

Some mothers are enrolled with another insurer at some point during pregnancy (29% of commercial and 9% of Medicaid mothers), so they may have an opportunity to strategically switch from the other insurer. However, there is little evidence of such switching. Few mothers (3% of commercial and 5% of Medicaid mothers) switch away from a less desirable insurer, i.e., an insurer with higher average out-of-pocket spending for OB care at the chosen OB hospital than the birth insurer, or an insurer where the chosen OB hospital is out-of-network. Most switching is between equal-out-of-pocket-spending insurers, and in fact 7% of commercial mothers switch away from lower-out-of-pocket-spending insurers. Thus, any insurer switching is likely unrelated to OB care (e.g., administrative churn).

B.5 Descriptive Evidence About Prenatal Care Providers

For each birth from October 9 to December 31, I use medical claims since the start of pregnancy (280 days before birth) to identify the prenatal care provider by NPI. Then, I use NPIs from

Table B5: Ruling Out Ways Home Location May Relate to Unobserved Shocks

	Attribute of Nearest Option			
Proxy for Unobserved Shocks	Distance (mi)	UAMS	NICU	ln(Non-OB Beds)
Conditional on $\mathbf{x}_i^{\text{demand}}$				
Married	0.063	−0.071	−0.058	0.006
	(0.647)	(0.072)	(0.045)	(0.085)
High School	−0.249	0.027	−0.005	0.009
	(0.334)	(0.034)	(0.035)	(0.036)
Hepatitis C	−0.584	−0.056	0.054	0.088
	(0.729)	(0.073)	(0.086)	(0.100)
Height (per SD, 2.8 in)	−0.023	0.010	−0.003	0.010
	(0.101)	(0.012)	(0.011)	(0.012)
NO_2 (per SD, 1.5 ppb)	0.035	0.190	0.167	0.119
	(1.735)	(0.201)	(0.121)	(0.214)
Conditional on $\mathbf{x}_i^{\text{health}}$				
Married	−0.069	−0.001	−0.002	0.012
	(0.169)	(0.006)	(0.013)	(0.017)
High School	−0.206	−0.003	0.005	0.020
	(0.264)	(0.011)	(0.017)	(0.022)
Hepatitis C	−0.119	−0.042	0.017	−0.026
	(0.684)	(0.057)	(0.049)	(0.055)
Height (per SD, 2.8 in)	0.030	0.001	−0.001	0.004
	(0.055)	(0.002)	(0.003)	(0.006)
NO_2 (per SD, 1.5 ppb)	0.225	0.122	0.041	0.029
	(1.545)	(0.139)	(0.110)	(0.156)
Observations	19843			

Sample: births with claims data, excluding outside option births. Attributes of nearest options are regressed with OLS on proxies for unobserved preference or underlying health shocks, controlling for mothers' characteristics that appear in either the demand or health model. Coefficients on proxies are reported. County-clustered SEs.

birth claims to determine at which OB hospitals those providers are practicing obstetricians. Summary statistics about prenatal care providers are shown in Table B8.

Most prenatal care providers are obstetricians, and they usually practice at OB hospitals near their offices. 83% of mothers deliver at the hospital where their prenatal care provider practices. Thus, the choice of OB hospital is effectively the joint choice of a prenatal care provider and hospital. This interpretation is consistent with the demand and health model: part of an OB hospital's characteristics is its set of prenatal care providers. For example, the demand and health effects of proximity to a prenatal care provider are included in the demand and health effects of proximity to an OB hospital.

This interpretation requires one assumption: in counterfactuals where an OB hospital

Table B6: Insurers Available to Each Mother

Payer Type	Births	Insurers
Medicaid	10526	1
ARHOME	4776	23
Employer-Sponsored	2406	35 (avg 1.3 in group)
State Employee	1262	7
ACA Marketplace	1117	19
Total	20087	85

Sample: births with claims data. Assumptions: mothers only choose within payer type, and mothers with employer-sponsored insurance only choose within group.

Table B7: Insurers Are Chosen Before, and Not During, Pregnancy

	Commercial	Medicaid
Had Birth Insurer Before Pregnancy (%)	73.6	61.7
Any Other Insurer During Pregnancy (%)	28.7	9.1
With Higher OOP (%)	3.4	5.2
With Equal OOP (%)	19.1	4.2
With Lower OOP (%)	7.1	0
Observations	2216	2368

Sample: births with claims data in Oct 9–Dec 31, excluding outside option births. Pregnancy length: 280 days. Higher OOP: other insurer has higher average OOP at chosen OB hospital than birth insurer, or chosen OB hospital is out-of-network for other insurer. Equal OOP: other insurer has equal average OOP as birth insurer. Lower OOP: other insurer has lower average OOP than birth insurer.

closes, mothers who travel farther to other OB hospitals also travel farther to prenatal care providers. This is plausible if OB unit closures force prenatal care providers to relocate their offices. I present descriptive evidence suggesting that such relocation is common.

I assign each prenatal care provider to the OB hospital where most of their births are delivered. Then, I examine whether that hospital is one of the four whose OB units closed in 2023–2024, and I examine whether each provider is based in a new location in 2025.[18] Table B9 shows the results. In 2025, 40% of closure-exposed prenatal care providers were located in a different city than they were in 2022, as opposed to 14% for all other prenatal care providers. This is consistent with prenatal care providers relocating to stay near OB hospitals.

B.6 Descriptive Evidence About Intensive-Margin Health Changes

I convert health effects to monetary units using the social cost of a preterm birth estimated by Waitzman et al. (2021), relying on the assumption that the marginal counterfactual-induced

[18] Data on provider locations come from CMS's National Plan and Provider Enumeration System.

Table B8: Summary Statistics About Prenatal Care Providers

	Share (%)
Prenatal Care Providers (Observations: 366)	
Obstetrician	65.6
Of Obstetricians: Main Practicing Hospital Within 5 Miles of Office	90.3
Of Obstetricians: Only One Practicing Hospital	87.5
Births (Observations: 3956)	
Prenatal Care Provider Is Obstetrician Practicing at Delivery Hospital	83.0
Prenatal Care Provider Is Obstetrician Who Delivered Birth	53.1

Sample: births with claims data in Oct 9–Dec 31, excluding outside option births, with identified prenatal care providers. First panel: those prenatal care providers, matched to births they deliver in 2022. Labeled as obstetrician if there are any matched births. Main practicing hospital is OB hospital where they deliver the most births.

Table B9: Relocation of Prenatal Care Providers After OB Unit Closures

	Share (%)	
	Exposed to 2023–2024 Closure	All Other Providers
Located in Different City in 2025	40.0	13.8
Located in Different State in 2025	30.0	9.0
No Longer Registered in 2025	0.0	0.6
Observations	10	356

Data source: NPPES. Sample: prenatal care providers for births with claims data in Oct 9–Dec 31, excluding outside option births. Each assigned to OB hospital where most of their births are delivered. Closure-exposed if that hospital is one of the 4 that closed their OB units in 2023–2024.

adverse-outcome birth is similar to a preterm birth. This is valid as long as counterfactuals only change health on the extensive margin (adverse-outcome births becoming no-adverse-outcome births, and vice versa) and not the intensive margin (adverse-outcome births becoming more or less severe). For example, if a policy that reduces adverse-outcome births also happens to lessen the severity of existing adverse-outcome births, I would undervalue its health effect.

In this section, I present descriptive evidence suggesting that extensive-margin health changes are unrelated to intensive-margin health changes. I define an intensive-margin health change as the difference between infant death and any non-fatal adverse outcome. I focus on health changes driven by OB hospital choice; as in Section 2.3, I proxy for OB hospital choice using which OB hospital is a mother's nearest option.

First, I repeat the maternal and infant health exercise in Section 2.3: I estimate Equation 1 to obtain ϕ_j, hospital-specific nearest-option effects on overall health. Then, I estimate the same equation replacing the outcome with an indicator for no non-fatal adverse outcomes, obtaining ϕ_j^{nonfatal}, nearest-option effects on non-fatal health.

Infant death is too rare to reliably identify hospital-specific effects. Instead, I estimate

$$\text{NoInfantDeath}_i = \mathbb{1}\left(\tilde{\tau}'\mathbf{x}_i + \tilde{\tau}^{\text{Dist}}\text{DistNearest}_i + \tilde{\tau}^{\phi}\sum_{j\in J^{\text{OB}}_{m(i)}}\phi_j^{\text{nonfatal}}\,\mathbb{1}\left(j \text{ nearest to } i\right) + \widetilde{\omega}_i > 0\right),$$
$$\widetilde{\omega}_i \sim \text{N}\left(0,1\right) \tag{B1}$$

$\tilde{\tau}^{\phi}$, the coefficient on the nearest option's ϕ_j^{nonfatal}, reflects the relationship between non-fatal health and infant death. If changes in non-fatal adverse outcomes are also associated with changes in infant death, the coefficient is positive. If they are unrelated, the coefficient is zero. Table B10 shows the results. I convert estimates to average marginal effects, where the nearest-option effects are relative to UAMS.

Table B10: Non-Fatal Adverse Outcome Effects Unrelated to Infant Death Effects

	Avg ΔPr(No Adverse Outcome) (%)		
	Any	Non-Fatal	Infant Death
Distance to Nearest Option (mi)	0.013	0.010	0.007
	(0.019)	(0.016)	(0.009)
Nearest-Option Effects (ϕ_j and ϕ_j^{nonfatal})			
Standard Deviation Across Hospitals	0.566	0.673	
	(0.218)	(0.177)	
Nearest Option's ϕ_j^{nonfatal}			−0.162
			(0.140)
Corr($\phi_j, \phi_j^{\text{nonfatal}}$)	0.883		
Observations	19843		

Results from estimating Equation 1, non-fatal version of Equation 1, and Equation B1. Sample: births with claims data, excluding outside option births. Drop hospitals that are nobody's nearest option (2 hospitals). Average marginal effects. Reference hospital for nearest-option effects: UAMS. Model-based SEs, third column takes ϕ_j^{nonfatal} as given.

First, I find no significant nearest-distance effect on any measure of health. Second, I find that ϕ_j and ϕ_j^{nonfatal} are highly correlated, consistent with non-fatal adverse outcomes being the adverse outcomes that matter on the extensive margin. Third, I report $\tilde{\tau}^{\phi}$ as the change in no-infant-death rate, in percentage points, per percentage point of the nearest option's ϕ_j^{nonfatal}. I find that it is negative and statistically insignificant, far below what a one-for-one relationship between non-fatal adverse outcomes and infant death would imply. Instead, it suggests they are unrelated. Thus, the evidence supports valuing health effects according to the social cost of the marginal counterfactual-induced adverse-outcome birth, assuming the severity of existing adverse-outcome births does not change.

C Demand and Health: Additional Results

The following tables present additional results from estimation of the model of OB hospital demand and maternal and infant health. Table C1 shows the effects of distance. Table C2 shows the effects of hospital size and having a NICU. Table C3 shows the effects of UAMS (the only teaching hospital with an obstetrics and gynecology residency program), WCWH (the only women's hospital), and the outside option. Table C4 shows the demand effects of the patient reviews index and the health effects of the patient outcomes index. Table C5 shows the health effects of mothers' characteristics. Table C6 shows the joint distributions of unobserved shocks.

Table C1: Demand and Health Results: Distance

	Median ΔUtility ($)	Avg ΔPr(Healthy) (%)
Distance (mi)	−225.86	−0.02138
	(69.91)	(0.00922)
ΔEffect of Distance per ΔDistance	−18.85	
	(11.36)	
ΔEffect of Distance (mi) per Δx_i:		
Medicaid	−108.49	
	(43.05)	
ARHOME	−86.47	
	(38.52)	
Age (yr)	−0.01	−0.00056
	(1.75)	(0.00066)
BMI	1.73	0.00099
	(1.34)	(0.00054)
Prior Births>0	−41.16	0.01006
	(24.77)	(0.00978)
Prior C-sections>0	39.20	0.00004
	(25.33)	(0.01379)
2×Pop Density	−70.67	
	(30.10)	
	Avg ΔUtility ($)	
Distance (mi)	−937.19	
	(476.91)	
Observations	20087	

Sample: births with claims data. Utility effects are median marginal effects, except last row which is average marginal effect. Health effects are average marginal effects. Utility effects evaluated at chosen distances. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

D Effects of OB Provision: Additional Results

The following figures present additional results related to the effects of OB provision. Figure D1 shows heterogeneity in access-to-care and maternal and infant health effects. Figure D2 shows the distance effects of OB provision. Figure D3 shows the role of distance in the access-to-care and maternal and infant health effects. Figure D4 compares the net externalities of rural and urban OB hospitals.

E Stylized Dynamic Model of OB Provision

In Section 6, OB provision is modeled as a static decision: it depends on current-period profitability, ignoring past or future outcomes. A realistic dynamic model would differ in two ways. First, building a brand-new OB unit may require a one-time sunk cost (e.g., construction). A hospital may refrain from operating an OB unit because the sunk cost is too high, even if it would earn a positive per-period profit. Second, hospitals may be forward-looking. A hospital may operate an unprofitable OB unit, or not operate a potentially profitable OB unit, due to expectations about the future.

If a hospital operates an OB unit that is unprofitable in the current period, I understate its fixed cost. If a hospital does not operate an OB unit that would be profitable in the current period, I overstate its fixed cost. In this section, I present a stylized dynamic model to illustrate assumptions under which these violations do not happen, and I argue that they are reasonable in this setting.

Stylized Dynamic Model Suppose time periods are indexed by $t \in \{0, 1, 2, \dots\}$, and future payoffs are discounted at factor $\chi \in (0, 1)$. In period t, an OB unit at hospital j generates variable profit π_{jt} for its system. Operating an OB unit costs fixed cost FC_j every period. Opening a new OB unit costs a one-time sunk cost $\text{SC}_j \leq \overline{\text{SC}}$, and closing an OB unit is free. Hospital j makes OB provision decision $\text{Prov}_{jt} \in \{0, 1\}$ at every period t.

A key assumption is that variable profits decline steadily: $\pi_{j,t+1} = \pi_{jt} - g_j$ with $g_j > 0$ for all hospitals j and periods t. Then, there is no reason to delay opening an OB unit, so a hospital either opens an OB unit at period 0 or never opens one. After opening, an OB unit operates until its variable profit falls below its fixed cost, at which point it closes and never reopens.

Suppose the market is observed at period $T > 0$. There are three types of hospitals. First, hospital j may operate an OB unit. This implies

$$\pi_{jT} \geq \text{FC}_j \tag{E1}$$

Second, hospital j may have opened an OB unit at period 0 but since closed it. The fact that it closed at or earlier than period T implies

$$\pi_{jT} \leq \text{FC}_j \tag{E2}$$

Third, hospital j may have never opened an OB unit. The fact that it did not open one at period 0 implies no possible sequence of decisions would make the sunk cost worth it. In particular, keeping the OB unit open until period T would not be worth it, so

$$\sum_{t=0}^{T} \chi^t \left(\pi_{jT} - \text{FC}_j + g_j (T-t)\right) \leq \text{SC}_j \tag{E3}$$

which implies

$$\pi_{jT} - \text{FC}_j \leq \frac{\text{SC}_j - g_j \sum_{t=0}^{T} \chi^t (T-t)}{\sum_{t=0}^{T} \chi^t} \tag{E4}$$

Because sunk cost SC_j is bounded above, T can be large enough that the right-hand side of Equation E4 is non-positive for all such hospitals j.

Thus, there exists a T' such that for all periods $T \geq T'$, all hospitals j satisfy

$$\text{Prov}_{jT} = \begin{cases} 1 & \text{if } \pi_{jT} > \text{FC}_j \\ 0 & \text{if } \pi_{jT} < \text{FC}_j \end{cases} \tag{E5}$$

This is exactly the static OB provision condition.

Discussion of Stylized Dynamic Model The stylized model illustrates two assumptions under which a static model captures all relevant features of OB provision. The first is that the profitability of OB provision is declining steadily, and the second is that it has been declining for a long time. Both assumptions are reasonable in this setting in 2022 given the history of OB provision, as shown in Appendix B.1. Between 1995 and 2024, there were zero openings of OB units and 24 closures of OB units in existing hospitals. This is consistent with the stylized model's prediction that a hospital either opens an OB unit at its first opportunity or never opens one, and an OB unit closes and never reopens after its per-period profit turns negative.

F Policies to Prevent OB Unit Closures: Additional Results

The following tables present additional results related to the evaluation of two policies, the Healthy Moms, Healthy Babies Act and an OB unit subsidy. Table F1 decomposes the policies'

effects on the individual components of social welfare. Tables F2, F3, and F4 show heterogeneity in the policies' access-to-care and maternal and infant health effects. Table F5 shows the policies' distance effects. Table F6 shows the role of distance in the policies' access-to-care and maternal and infant health effects. Table F7 shows OB unit closures among rural and urban hospitals.

G Sensitivity of Results to Discounting Mothers' Welfare

The access-to-care effect is a change in mothers' demand-based welfare, which rests on revealed preferences. A potential concern is that those preferences reflect choice frictions that are not welfare-relevant. For example, a mother may stick with a hospital or prenatal care provider chosen for reasons unrelated to pregnancy due to inertia, or a mother may have imperfect information about distant hospitals or out-of-pocket prices. If these frictions are important, then the access-to-care effect is overstated.

In this section, I measure how much the access-to-care effect can be discounted to account for choice frictions before the main conclusions of this paper change. I consider scaling mothers' welfare by a factor between 0 and 1 and recomputing the net externality of each OB hospital (as in Section 5) and the marginal values of public funds of the two policies (as in Section 7). I also consider scaling only the part of mothers' welfare explained by expected travel distance, as there may be choice frictions specifically related to distance preferences.

Table G1 shows the results. The average OB hospital's net externality remains positive until mothers' welfare is scaled by 0.36, and both policies' marginal values of public funds remain above 1 until mothers' welfare is scaled by 0.40. Even with the distance-related part of mothers' welfare set to zero, the average net externality is positive, and both marginal values of public funds remain above 1.

Table C2: Demand and Health Results: Hospital Size and NICU

	Avg ΔUtility ($)	Avg ΔPr(Healthy) (%)
2×Non-OB Beds	3638.0	−0.187
	(2198.3)	(0.383)
ΔEffect of 2×Non-OB Beds per Δx_i:		
Medicaid	−261.3	
	(357.4)	
ARHOME	−166.4	
	(345.2)	
Age (yr)	119.0	−0.001
	(33.1)	(0.014)
BMI	59.0	0.024
	(15.7)	(0.013)
Prior Births>0	−521.1	0.168
	(156.0)	(0.179)
Prior C-sections>0	633.3	0.123
	(376.9)	(0.288)
2×Pop Density	350.3	
	(129.1)	
NICU	−1014.4	0.762
	(2746.2)	(1.122)
ΔEffect of NICU per Δx_i:		
Medicaid	−1117.8	
	(900.3)	
ARHOME	−1301.8	
	(704.1)	
Age (yr)	−32.6	0.069
	(31.6)	(0.060)
BMI	−73.9	0.003
	(33.6)	(0.033)
Prior Births>0	−132.7	−0.335
	(288.0)	(0.759)
Prior C-sections>0	−978.6	−0.283
	(582.5)	(0.867)
2×Pop Density	99.5	
	(207.0)	
Observations	20087	

Sample: births with claims data. Average marginal effects. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

Table C3: Demand and Health Results: UAMS, WCWH, and Outside Option

	Avg ΔUtility ($)	Avg ΔPr(Healthy) (%)
UAMS	−2993.0	0.247
	(5811.2)	(1.378)
ΔEffect of UAMS per Δx_i:		
Medicaid	5163.3	
	(1761.2)	
ARHOME	2610.8	
	(902.7)	
Age (yr)	74.1	−0.086
	(34.1)	(0.058)
BMI	66.6	−0.104
	(28.3)	(0.040)
Prior Births>0	476.5	−1.495
	(389.0)	(0.655)
Prior C-sections>0	−4349.6	1.631
	(1607.5)	(1.198)
2×Pop Density	336.4	
	(196.6)	
WCWH	9643.4	
	(3726.0)	
Outside Option	−23239.0	−4.638
	(6573.9)	(2.415)
Observations	20087	

Sample: births with claims data. Average marginal effects. Effects relative to a hospital with same observed characteristics and zero quality shock. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

Table C4: Demand and Health Results: Patient Reviews and Patient Outcomes

	Avg ΔUtility ($)
Patient Reviews (1 SD)	−262.0
	(1566.6)
ΔEffect of Patient Reviews (1 SD) per Δx_i:	
Medicaid	−1119.4
	(583.4)
ARHOME	−1118.1
	(577.8)
Age (yr)	28.8
	(18.3)
BMI	40.1
	(15.8)
Prior Births>0	−852.5
	(334.4)
Prior C-sections>0	137.1
	(256.2)
2×Pop Density	−182.4
	(137.6)
	Avg ΔPr(Healthy) (%)
Patient Outcomes (1 SD)	−0.200
	(0.244)
ΔEffect of Patient Outcomes (1 SD) per Δx_i:	
Age (yr)	−0.004
	(0.016)
BMI	0.004
	(0.013)
Prior Births>0	0.519
	(0.257)
Prior C-sections>0	0.105
	(0.362)
Observations	20087

Sample: births with claims data. Average marginal effects. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

Table C5: Demand and Health Results: Mothers' Characteristics in Health

	Avg ΔPr(Healthy) (%)		Avg ΔPr(Healthy) (%)
Medicaid	0.072	Race (vs. Asian):	
	(0.502)	Black	−0.233
ARHOME	−0.039		(0.876)
	(0.532)	Native American	1.588
Age (yr)	−0.028		(1.341)
	(0.032)	Pacific Islander	0.529
BMI	0.001		(0.691)
	(0.016)	Race Unknown	1.551
Prior Births>0	1.727		(0.774)
	(0.346)	White	1.193
Prior C-sections>0	−0.521		(0.818)
	(0.346)	Income Share (%), by % of Poverty Line:	
Diabetes	−2.300	<50%	0.150
	(0.867)		(0.088)
Hypertension	−0.067	50–99%	−0.192
	(0.688)		(0.079)
Smoker	−0.840	100–124%	0.135
	(0.363)		(0.113)
2×Pop Density	−0.096	125–149%	0.108
	(0.127)		(0.109)
		150–184%	0.130
			(0.105)
		185–199%	−0.294
			(0.181)
Observations	20087		

Sample: births with claims data. Average marginal effects. County-clustered SEs.

Table C6: Demand and Health Results: Unobserved Shocks

Preferences	ΔEffect per SD of ν_i^k (\$)	Corr(ν_i^k, ϵ_i)
Distance (mi)	−835.1	0.06
	(330.0)	(0.05)
UAMS	7757.9	−0.33
	(3027.0)	(0.18)
NICU	6477.1	−0.04
	(2369.4)	(0.10)
2×Non-OB Beds	2688.0	−0.03
	(912.4)	(0.11)
Patient Reviews (1 SD)	4402.9	−0.01
	(1404.8)	(0.07)
	SD of ξ_j (\$)	SD of λ_j (%)
Demand-Based and Health-Based Quality	1880.8	1.04
	(476.8)	(0.24)
Corr(ξ_j, λ_j)		0.62
		(0.37)
Observations	20087	

Sample: births with claims data. Effect of distance preference shock is median marginal effect. For distance, a bigger shock is negative preference. For other characteristics, a bigger shock is positive preference. County-clustered SEs for estimates related to mother-level variation, model-based SEs for estimates related to only hospital-level variation.

Table F1: Healthy Moms, Healthy Babies Act and OB Subsidy: Decomposition of Social Welfare Effects

	ΔMothers' Welfare (\$1m)	ΔValue of Healthy Births (\$1m)	Variable Cost Savings (\$1m)	Fixed Cost Savings (\$1m)
\$100k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+1.2, +18.2]	[+0.1, +0.8]	[+0.002, +1.3]	[−8.8, −0.5]
+ Medicaid Price +70%	[+1.4, +18.4]	[+0.1, +0.8]	[−0.03, +1.3]	[−8.9, −0.6]
+ OB Subsidy \$100k	[+1.4, +18.4]	[+0.1, +0.8]	[−0.04, +1.2]	[−8.9, −0.5]
\$200k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+2.0, +12.8]	[+0.1, +0.6]	[−2.4, +0.7]	[−5.6, −0.8]
+ Medicaid Price +70%	[+2.5, +22.3]	[+0.1, +0.8]	[−0.005, +0.9]	[−10.2, −0.9]
+ OB Subsidy \$100k	[+1.4, +6.2]	[+0.01, +0.3]	[−0.9, +0.2]	[−2.6, −0.5]
+ OB Subsidy \$200k	[+2.8, +22.7]	[+0.2, +0.8]	[−0.1, +0.8]	[−10.4, −1.0]
\$300k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+2.7, +11.7]	[+0.1, +0.7]	[−1.0, +0.2]	[−4.9, −1.0]
+ Medicaid Price +70%	[+3.8, +24.0]	[+0.2, +1.3]	[+0.1, +1.0]	[−10.3, −1.4]
+ OB Subsidy \$100k	[+1.7, +5.2]	[+0.1, +0.7]	[−0.02, +0.9]	[−2.0, −0.5]
+ OB Subsidy \$200k	[+3.1, +11.5]	[+0.2, +0.7]	[−0.1, +0.3]	[−4.6, −1.0]
+ OB Subsidy \$300k	[+4.5, +27.9]	[+0.3, +1.5]	[−0.1, +1.2]	[−11.6, −1.6]

Actual policy is 70% Medicaid price hike. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcomes are mothers' demand-based welfare before OOP, value of expected no-adverse-outcome births, expected variable cost savings, and fixed cost savings.

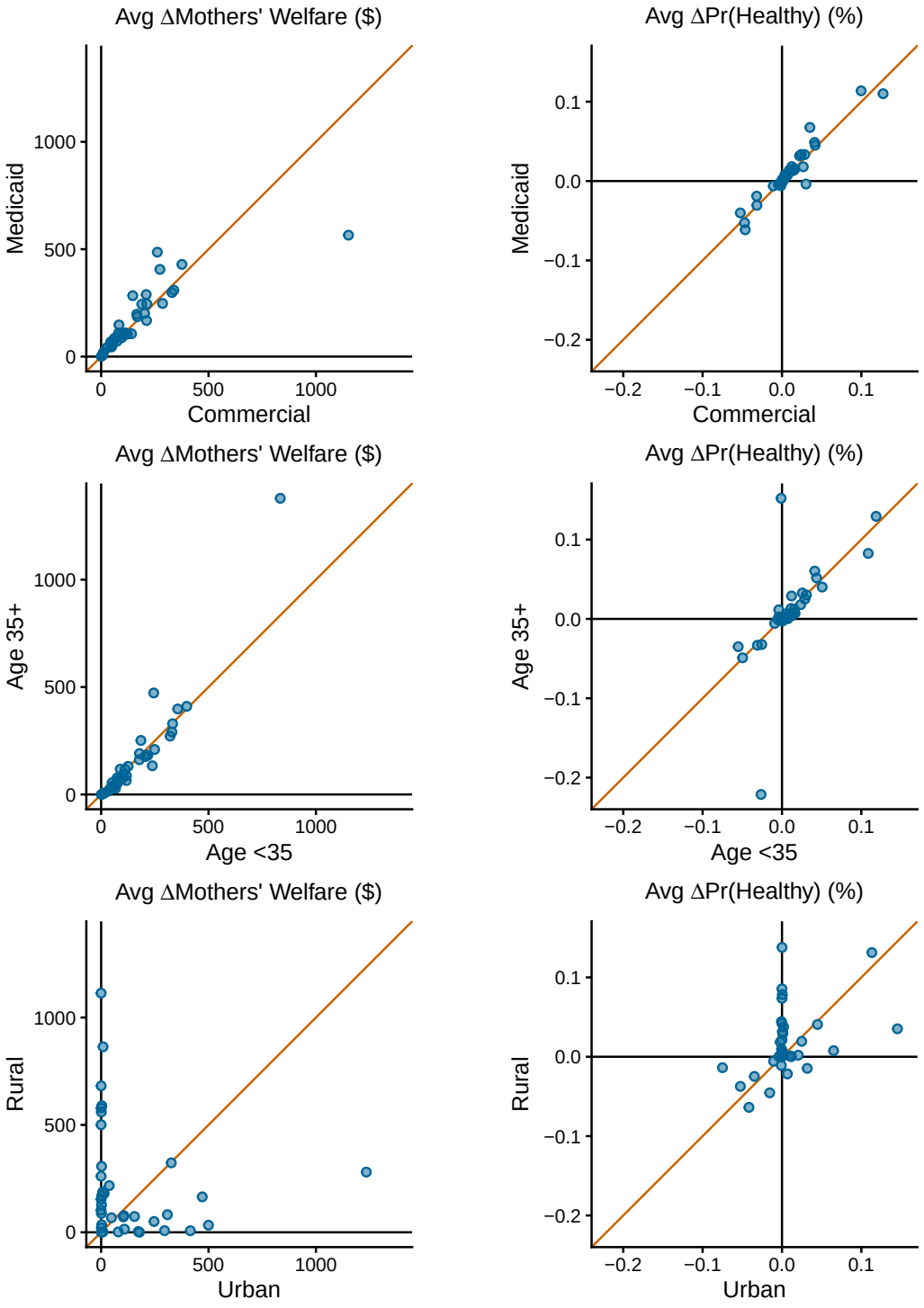


Figure D1: Heterogeneity in Access-to-Care and Maternal and Infant Health Effects Between Types of Mothers

Sample: OB hospitals. Axes are components of variable social surplus, divided by births: change in mothers' demand-based welfare before OOP, and change in expected no-adverse-outcome births caused by hospital's OB provision. Sloped line is 45° line.

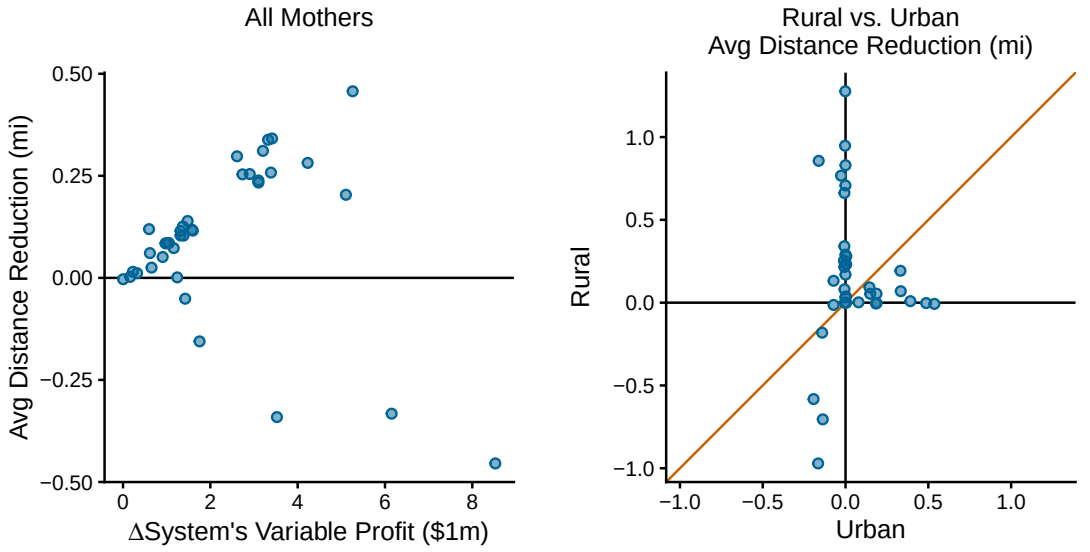


Figure D2: Distance Effects of OB Provision

Sample: OB hospitals. Horizontal axis in left plot is change in system's expected variable profit caused by hospital's OB provision. Vertical axis in left plot, and both axes in right plot, are reduction in expected travel distance caused by hospital's OB provision, divided by births. Sloped line is 45° line.

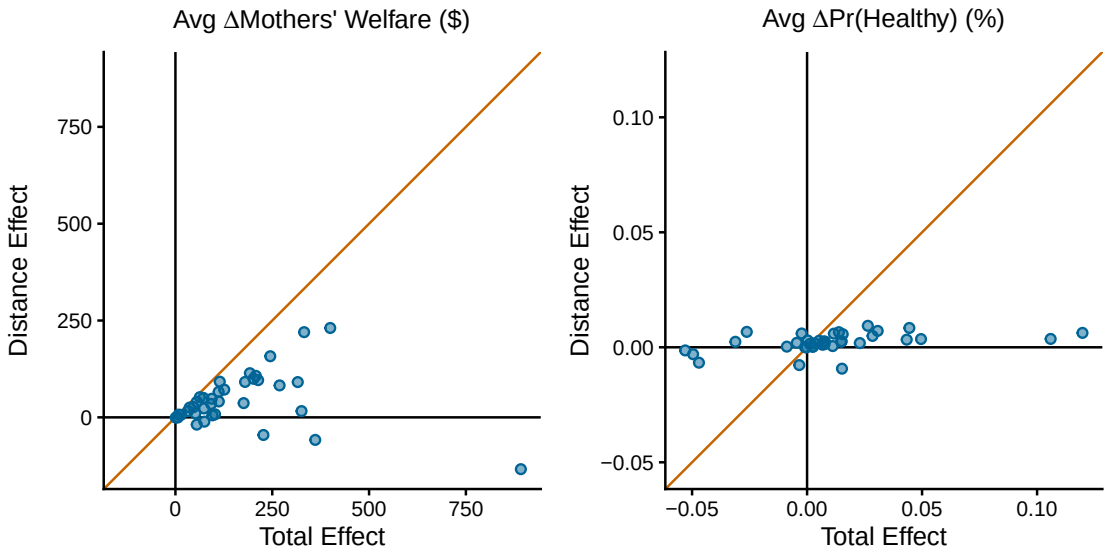


Figure D3: Access-to-Care and Maternal and Infant Health Effects, Separating Distance Effect From Total Effect

Sample: OB hospitals. Axes are components of variable social surplus, divided by births: change in mothers' demand-based welfare before OOP, and change in expected no-adverse-outcome births caused by hospital's OB provision. Distance effect is part explained only by reduction in expected travel distance. Sloped line is 45° line.

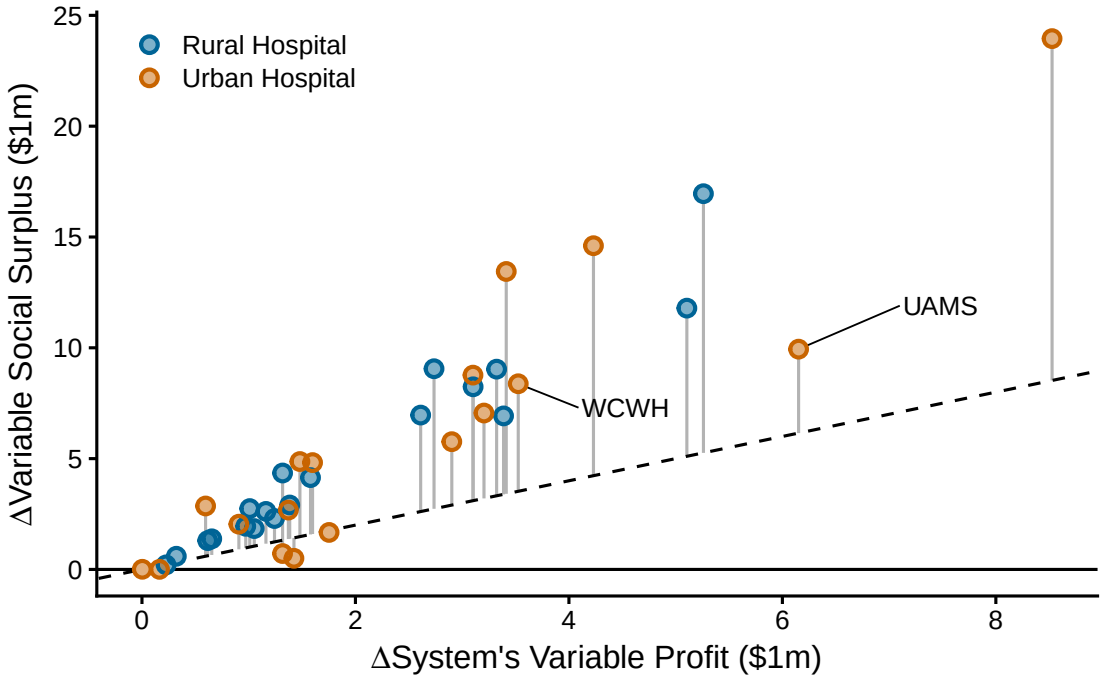


Figure D4: Net Externality of OB Provision, Rural and Urban Hospitals

Sample: OB hospitals. Horizontal axis is change in system's expected variable profit caused by hospital's OB provision. Vertical axis is variable social surplus, summing the following: change in mothers' demand-based welfare before OOP, expected variable cost savings, and value of change in expected no-adverse-outcome births caused by hospital's OB provision. Sloped line is 45° line, and vertical lines denote net externalities. Labeled hospitals are UAMS (teaching) and WCWH (women's).

Table F2: Healthy Moms, Healthy Babies Act and OB Subsidy: Heterogeneity in Effects Between Medicaid and Commercial Mothers

	Avg ΔMothers' Welfare ($)		Avg ΔPr(Healthy) (%)	
	Medicaid	Commercial	Medicaid	Commercial
$100k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+42, +584]	[+35, +545]	[+0.004, +0.041]	[+0.003, +0.030]
+ Medicaid Price +70%	[+46, +591]	[+39, +551]	[+0.005, +0.042]	[+0.003, +0.030]
+ OB Subsidy $100k	[+47, +592]	[+39, +551]	[+0.004, +0.041]	[+0.003, +0.030]
$200k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+70, +434]	[+58, +364]	[+0.005, +0.033]	[+0.001, +0.021]
+ Medicaid Price +70%	[+84, +745]	[+71, +649]	[+0.007, +0.042]	[+0.005, +0.029]
+ OB Subsidy $100k	[+47, +212]	[+39, +179]	[+0.002, +0.017]	[−0.001, +0.011]
+ OB Subsidy $200k	[+94, +759]	[+78, +660]	[+0.008, +0.043]	[+0.005, +0.029]
$300k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+92, +395]	[+79, +338]	[+0.007, +0.039]	[+0.005, +0.025]
+ Medicaid Price +70%	[+129, +846]	[+109, +664]	[+0.011, +0.069]	[+0.008, +0.045]
+ OB Subsidy $100k	[+59, +205]	[+49, +151]	[+0.006, +0.038]	[+0.005, +0.024]
+ OB Subsidy $200k	[+106, +386]	[+89, +330]	[+0.010, +0.039]	[+0.007, +0.023]
+ OB Subsidy $300k	[+153, +964]	[+128, +783]	[+0.014, +0.081]	[+0.010, +0.053]

Actual policy is 70% Medicaid price hike. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcomes are mothers' demand-based welfare before OOP, and expected no-adverse-outcome births, divided by births.

Table F3: Healthy Moms, Healthy Babies Act and OB Subsidy: Heterogeneity in Effects Between Older and Younger Mothers

	Avg ΔMothers' Welfare ($)		Avg ΔPr(Healthy) (%)	
	Age 35+	Age <35	Age 35+	Age <35
$100k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+30, +476]	[+39, +573]	[+0.002, +0.022]	[+0.004, +0.036]
+ Medicaid Price +70%	[+32, +480]	[+43, +579]	[+0.003, +0.022]	[+0.004, +0.037]
+ OB Subsidy $100k	[+32, +480]	[+44, +580]	[+0.002, +0.022]	[+0.004, +0.036]
$200k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+51, +344]	[+65, +401]	[+0.003, +0.021]	[+0.004, +0.027]
+ Medicaid Price +70%	[+59, +599]	[+79, +702]	[+0.004, +0.025]	[+0.006, +0.036]
+ OB Subsidy $100k	[+31, +164]	[+44, +197]	[+0.001, +0.010]	[−0.00001, +0.014]
+ OB Subsidy $200k	[+63, +605]	[+88, +715]	[+0.004, +0.026]	[+0.007, +0.036]
$300k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+72, +330]	[+86, +367]	[+0.004, +0.032]	[+0.006, +0.031]
+ Medicaid Price +70%	[+95, +663]	[+120, +754]	[+0.006, +0.043]	[+0.010, +0.057]
+ OB Subsidy $100k	[+42, +142]	[+55, +164]	[+0.003, +0.023]	[+0.006, +0.031]
+ OB Subsidy $200k	[+73, +302]	[+99, +361]	[+0.005, +0.027]	[+0.009, +0.031]
+ OB Subsidy $300k	[+105, +748]	[+143, +877]	[+0.007, +0.049]	[+0.012, +0.067]

Actual policy is 70% Medicaid price hike. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcomes are mothers' demand-based welfare before OOP, and expected no-adverse-outcome births, divided by births.

Table F4: Healthy Moms, Healthy Babies Act and OB Subsidy: Heterogeneity in Effects Between Rural and Urban Mothers

	Avg ΔMothers' Welfare ($)		Avg ΔPr(Healthy) (%)	
	Rural	Urban	Rural	Urban
$100k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+55, +1107]	[+29, +257]	[+0.007, +0.086]	[+0.002, +0.006]
+ Medicaid Price +70%	[+62, +1121]	[+30, +259]	[+0.008, +0.088]	[+0.001, +0.006]
+ OB Subsidy $100k	[+62, +1121]	[+32, +261]	[+0.008, +0.088]	[+0.001, +0.005]
$200k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+87, +727]	[+50, +209]	[+0.009, +0.066]	[−0.008, +0.008]
+ Medicaid Price +70%	[+110, +1325]	[+58, +336]	[+0.013, +0.092]	[+0.002, +0.009]
+ OB Subsidy $100k	[+63, +350]	[+31, +106]	[+0.007, +0.033]	[−0.003, +0.003]
+ OB Subsidy $200k	[+125, +1350]	[+63, +342]	[+0.016, +0.095]	[+0.001, +0.008]
$300k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+108, +588]	[+71, +238]	[+0.011, +0.075]	[+0.003, +0.011]
+ Medicaid Price +70%	[+165, +1560]	[+91, +327]	[+0.019, +0.155]	[−0.0001, +0.012]
+ OB Subsidy $100k	[+81, +301]	[+38, +92]	[+0.011, +0.073]	[+0.002, +0.006]
+ OB Subsidy $200k	[+144, +635]	[+69, +198]	[+0.019, +0.080]	[+0.002, +0.008]
+ OB Subsidy $300k	[+207, +1651]	[+101, +422]	[+0.027, +0.168]	[+0.003, +0.012]

Actual policy is 70% Medicaid price hike. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcomes are mothers' demand-based welfare before OOP, and expected no-adverse-outcome births, divided by births.

Table F5: Healthy Moms, Healthy Babies Act and OB Subsidy: Distance Effects

	Avg Distance Reduction (mi)		
	All Mothers	Rural	Urban
$100k Fixed Cost Increase	–	–	–
+ Medicaid Price +35%	[+0.04, +0.53]	[+0.06, +1.22]	[+0.03, +0.15]
+ Medicaid Price +70%	[+0.04, +0.54]	[+0.07, +1.24]	[+0.03, +0.15]
+ OB Subsidy $100k	[+0.04, +0.54]	[+0.07, +1.24]	[+0.02, +0.14]
$200k Fixed Cost Increase	–	–	–
+ Medicaid Price +35%	[+0.06, +0.38]	[+0.09, +0.80]	[+0.04, +0.14]
+ Medicaid Price +70%	[+0.08, +0.64]	[+0.12, +1.41]	[+0.05, +0.20]
+ OB Subsidy $100k	[+0.04, +0.19]	[+0.07, +0.39]	[+0.02, +0.07]
+ OB Subsidy $200k	[+0.08, +0.65]	[+0.14, +1.45]	[+0.04, +0.20]
$300k Fixed Cost Increase	–	–	–
+ Medicaid Price +35%	[+0.08, +0.33]	[+0.12, +0.62]	[+0.06, +0.19]
+ Medicaid Price +70%	[+0.12, +0.73]	[+0.19, +1.73]	[+0.08, +0.23]
+ OB Subsidy $100k	[+0.06, +0.21]	[+0.10, +0.40]	[+0.03, +0.10]
+ OB Subsidy $200k	[+0.10, +0.34]	[+0.17, +0.71]	[+0.06, +0.16]
+ OB Subsidy $300k	[+0.14, +0.85]	[+0.24, +1.85]	[+0.08, +0.30]

Actual policy is 70% Medicaid price hike. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcome is reduction in expected travel distance, divided by births.

Table F6: Healthy Moms, Healthy Babies Act and OB Subsidy: Access-to-Care and Maternal and Infant Health Effects, Separating Distance Effect From Total Effect

	Avg ΔMothers' Welfare ($)		Avg ΔPr(Healthy) (%)	
	Total Effect	Distance Effect	Total Effect	Distance Effect
$100k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+38, +562]	[+19, +262]	[+0.003, +0.035]	[+0.001, +0.012]
+ Medicaid Price +70%	[+42, +569]	[+21, +265]	[+0.004, +0.035]	[+0.001, +0.012]
+ OB Subsidy $100k	[+43, +569]	[+21, +264]	[+0.004, +0.035]	[+0.001, +0.012]
$200k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+63, +395]	[+27, +175]	[+0.004, +0.027]	[+0.001, +0.008]
+ Medicaid Price +70%	[+77, +691]	[+38, +283]	[+0.006, +0.035]	[+0.002, +0.015]
+ OB Subsidy $100k	[+43, +193]	[+20, +86]	[+0.0004, +0.014]	[+0.001, +0.004]
+ OB Subsidy $200k	[+85, +704]	[+41, +287]	[+0.007, +0.035]	[+0.002, +0.015]
$300k Fixed Cost Increase	–	–	–	–
+ Medicaid Price +35%	[+84, +364]	[+39, +143]	[+0.006, +0.031]	[+0.002, +0.007]
+ Medicaid Price +70%	[+118, +744]	[+59, +291]	[+0.009, +0.056]	[+0.003, +0.016]
+ OB Subsidy $100k	[+54, +161]	[+27, +86]	[+0.005, +0.030]	[+0.001, +0.004]
+ OB Subsidy $200k	[+96, +355]	[+47, +153]	[+0.008, +0.030]	[+0.002, +0.008]
+ OB Subsidy $300k	[+139, +863]	[+68, +374]	[+0.012, +0.065]	[+0.003, +0.019]

Actual policy is 70% Medicaid price hike. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost. Outcomes are mothers' demand-based welfare before OOP, and expected no-adverse-outcome births, each divided by births. Distance effect is part explained only by reduction in expected travel distance.

Table F7: Healthy Moms, Healthy Babies Act and OB Subsidy: Rural and Urban Hospitals

	OB Closures	
	Rural Hospitals (Out of 19)	Urban Hospitals (Out of 18)
$100k Fixed Cost Increase	[0.6, 3.6]	[1.5, 2.5]
+ Medicaid Price +35%	[0.2, 0.3]	[1.1, 1.2]
+ Medicaid Price +70%	0	1
+ OB Subsidy $100k	0	0
$200k Fixed Cost Increase	[1.2, 4.5]	[2.1, 3.3]
+ Medicaid Price +35%	[0.7, 3.1]	[1.6, 2.5]
+ Medicaid Price +70%	[0.3, 0.5]	[1.3, 1.4]
+ OB Subsidy $100k	[0.6, 3.6]	[1.5, 2.5]
+ OB Subsidy $200k	0	0
$300k Fixed Cost Increase	[2.0, 5.5]	[2.5, 4.2]
+ Medicaid Price +35%	[1.3, 4.6]	[2.1, 2.9]
+ Medicaid Price +70%	[0.8, 1.3]	[1.8, 2.4]
+ OB Subsidy $100k	[1.2, 4.5]	[2.1, 3.3]
+ OB Subsidy $200k	[0.6, 3.6]	[1.5, 2.5]
+ OB Subsidy $300k	0	0

Actual policy is 70% Medicaid price hike. Reported values are predicted outcomes: averages over fixed cost shocks drawn from uniform posterior. I report the range of predicted outcomes across the estimated set of expected fixed cost.

Table G1: Sensitivity of Results to Discounting Mothers' Welfare

Scale	Average Net Externality ($1m)	MVPF	
		Medicaid Price Hike	OB Unit Subsidy
All Mothers' Welfare			
1.00	3.37	[1.03, 1.54]	[1.24, 4.17]
0.75	2.06	[1.02, 1.33]	[1.15, 2.91]
0.50	0.76	[1.01, 1.13]	[1.05, 1.66]
0.25	−0.55	[0.88, 1.00]	[0.41, 0.97]
Distance Part Only			
0.50	2.63	[1.02, 1.38]	[1.15, 3.00]
0.00	1.88	[1.01, 1.22]	[1.06, 1.84]

First panel: mothers' demand-based welfare before out-of-pocket spending is multiplied by the scale. Second panel: only the part explained by expected travel distance is scaled. MVPF: range across all fixed cost scenarios and all values of expected fixed cost in the estimated set, for the 70% Medicaid price hike and fully offsetting OB unit subsidy.